Beyond the Jack-of-All-Trades Hypothesis: The Importance of Business and Specialized Skills for Business Ownership Entry and Success

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Abstract

What types of skills does a worker need to start and run a business? Lazear's (2004, 2005) jack-of-all-trades (JOAT) hypothesis holds that workers with balanced skills are more likely to become business owners, and empirical studies support it, typically by showing that workers who have held more occupations or employers are more likely to do so. I argue that what prepares a worker for business ownership is not balance across all skills but two types of skills: business skills and specialized skills relevant to the business. I also distinguish between basic and advanced business ownership, where advanced businesses are incorporated and/or start with more capital. Finally, I build a direct measure of skill balance and compare it with the conventional occupation count. Using NLSY79 work histories linked to O*NET, I find that business skills predict entry into advanced business ownership, by 34 percent per half standard deviation, while directly measured skill balance does not predict entry into either type. Distinguishing the two types of business ownership also shows that they reward different skills. Business skills are more strongly associated with business revenue and earnings in advanced businesses, while specialized skills are more strongly associated with them in basic businesses. Unlike my direct measure of skill balance, the occupation count predicts entry only into basic business ownership, partly because it reflects involuntary job losses. Owners of advanced businesses are jacks of *business* trades rather than jacks of all trades.

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1 Introduction

What kind of work experience prepares someone to start a business? Arguably, the best-known answer comes from Lazear's (2004, 2005) jack-of-all-trades (JOAT) model, which posits that because entrepreneurs must handle many functions at once, such as marketing, finance, and operations, workers with balanced skills have a comparative advantage in entrepreneurship, while workers with specialized skills have a comparative advantage in wage employment.[1] An extensive empirical literature finds largely supportive evidence for this hypothesis, typically by using regression analysis to relate a measure of skill balance to the probability of becoming an entrepreneur (Wagner, 2003, 2006; Stuetzer, Obschonka and Schmitt-Rodermund, 2013; Lechmann and Schnabel, 2014; Åstebro and Thompson, 2011; Aldén, Hammarstedt and Neuman, 2017). In these studies, entrepreneurship is defined as any self-employment and/or business ownership,[2] and skill balance is usually proxied by the number of occupations or employers a worker has held, without distinguishing between the skills different experiences develop. Yet Lazear's own example of a founder suggests a more nuanced view of the skills needed for business ownership: "Consider the founder of a new small restaurant. In addition to being a good cook, the founder must be able to obtain funds, hire workers, choose location and decor, obtain food supplies at a reasonable cost, keep books, and market the restaurant" (Lazear, 2004, p. 208). He implicitly suggests that workers need business skills and specialized (cooking) skills. However, this distinction is lost in subsequent empirical work, including Lazear's (2005) own test.

I extend this literature by introducing three innovations. First, I argue that what prepares a worker for business ownership is not balance across *all* skills, but rather two skill types relevant to the establishment and operation of a business: *business skills*, meaning core business knowledge, managerial skills, and interpersonal skills, and *specialized skills*, meaning expertise directly relevant to the business. This distinction matters because conventional measures of skill balance, which count the number of occupations or employers a worker has held, treat experience across different occupations or employers as equally informative about a worker's preparedness for business ownership. For example, these measures would suggest that a worker who was a truck driver, a chef, and a manager before opening a restaurant is better equipped to do so than a worker who spent the same amount of time as a chef

[1] Even those entrepreneurs who put together teams to launch a business require balanced skills to select and manage the team.

[2] Although Lazear (2004, 2005) uses the term "entrepreneur" in his work, he has in mind owners of relatively complex businesses (including, for example, restaurants) who may or may not be innovators as envisioned by Schumpeter (1912) in his widely used definition of an entrepreneur. I follow the majority of JOAT-related studies in considering business owners to be anyone working for their own business; because these workers are reported in my data as either self-employed or business owners or both (see Section 4), I use the terms "self-employment" and "business ownership" interchangeably throughout this paper.

and a manager, even though truck driving is unlikely to develop skills relevant to running a restaurant. By measuring business skills and specialized skills directly, I focus on the types of skills most relevant to business ownership.

Second, I distinguish between two types of business ownership. Most studies pool all self-employment and/or business ownership, which means, for example, that a freelance house cleaner and the founder of an incorporated construction firm fall into the same category even though they perform different functions and may require different skills. Both the JOAT hypothesis and my argument that business skills are what matter are likely to apply to businesses that are complex enough to require coordinating multiple functions, such as the restaurant in Lazear's example. Pooling these businesses with simpler businesses can obscure the roles that different skills play. I classify businesses that are incorporated and/or start with higher levels of capital as *advanced* businesses, where owners are more likely to manage functions such as financing, hiring, and marketing in addition to using their specialized skills; all other businesses are classified as *basic* businesses, where owners are expected to rely primarily on specialized skills. My definition of advanced businesses appears to conform to Lazear's (2004, 2005) definition of an "entrepreneur," as discussed in footnote 2.

Third, I revisit conventional tests of the JOAT hypothesis by comparing the occupation count with a direct measure of skill balance. The occupation count can be a poor proxy for skill balance for two reasons. First, it treats every new occupation as an equal step toward balance, however similar the occupation is to those a worker has held before. For example, a worker who moves from carpenter to cabinetmaker adds an occupation to the count, even though both jobs build much the same skills. Second, the count also rises with job churning, such as repeated involuntary job loss, which has little to do with skill accumulation. Because workers with unstable careers may turn to business ownership when their other options are limited, the count can predict entry into business ownership even if skill balance plays no role. By comparing the occupation count with a direct measure of skill balance, I can assess whether the occupation count predicts entry into business ownership because it captures skill balance or because it reflects job churning.

Before pursuing these three innovations empirically, I present a simple static choice model to clarify how business skills and specialized skills affect workers' choices among wage employment, basic business ownership, and advanced business ownership, as well as their success in each type of business. In the model, entry depends on how the returns to a skill in business ownership compare with its returns in wage employment, whereas success depends on its returns within business ownership, so a skill can discourage entry yet still raise success after entry. I then turn to an empirical analysis in which I combine decades-long work histories from the 1979 National Longitudinal Survey of Youth (NLSY79) with occupation-specific

skill data from the Occupational Information Network (O*NET). I use these data to estimate a competing-risks model of workers' year-by-year transitions from wage employment or nonemployment into either basic or advanced business ownership. I also examine subsequent success in each type of business, measured by business revenue and earnings. The key regressors are measures of business skills and specialized skills, which I construct by weighting the skill and knowledge ratings of each occupation a worker has held, based on O*NET, by the years spent in that occupation. I also construct a direct measure of skill balance based on how evenly a worker's accumulated skills are spread across different types of skills and knowledge. I use this measure, along with conventional occupation counts, in the business ownership entry model to replicate and extend tests of the JOAT hypothesis.[3]

My findings have three implications for the JOAT hypothesis. First, what matters for business ownership is not balance across all skills, but the types of skills workers accumulate. My direct measure of skill balance is unrelated to entry once business and specialized skills are accounted for, and shows little relationship to success. In contrast, a half-standard-deviation increase in business skills is predicted to increase the probability of entering advanced business ownership by 34.4 percent. This suggests that owners of advanced businesses are jacks of *business* trades rather than jacks of all trades. This finding is in line with Chen and Thompson (2016), who find that binary indicators for whether a worker has ever worked in specific areas, such as finance and senior management, largely explain the relationship between skill balance and founding a business. My results show that this pattern holds for a broad measure of business skills, which captures how much core business knowledge, managerial skills, and interpersonal skills a worker has accumulated throughout her career.

Second, distinguishing between types of business ownership reveals important differences in how business skills and specialized skills predict entry into and success in business ownership. Business skills predict higher entry into advanced business ownership but lower entry into basic business ownership, consistent with advanced businesses rewarding business skills more, and basic businesses less, than wage employment does. Specialized skills, by contrast, predict lower entry into both types. After entry, both types of skills predict higher business revenue and earnings, but their relative importance differs by business type. Business skills predict revenue and earnings more strongly in advanced business ownership, where a half-standard-deviation increase is associated with 52.0 percent higher revenue, while specialized

[3]I cannot completely separate a direct (causal) effect of business skills and specialized skills from the effects of other factors (e.g., wealth, risk tolerance, a preference for autonomy) that affect skill acquisition via occupational choice and also affect entry into business ownership and success outcomes. I use a "selection on observables" strategy by controlling for measures of wealth, risk tolerance, cognitive ability, personality traits, family background, and other factors. I describe this approach (as well as my inability to use instrumental variables estimation) in Section 5.

skills predict them more strongly in basic business ownership, where the same increase is associated with 17.6 percent higher revenue. The distinction between types of business ownership also matters for conventional tests of the JOAT hypothesis. Because basic business ownership is more common and accounts for much of the positive relationship between occupation counts and entry into business ownership, estimates that pool the two types may overstate support for the JOAT hypothesis.

Finally, the occupation count, often used as a proxy for skill balance, also reflects job churning. It predicts entry into basic business ownership only, and about a third of this relationship is explained by involuntary job loss. Together, these findings suggest that conventional tests of the JOAT hypothesis may conflate the role of skill balance with those of business skills and job churning, while obscuring important differences between types of business ownership.

These findings also have implications for policies that promote business ownership. They suggest that efforts to prepare prospective business owners should focus on building business skills, such as managerial, financial, and interpersonal skills, rather than a broad balance of skills. Programs supported by the U.S. Small Business Administration (SBA), such as Small Business Development Centers (SBDCs) and SCORE,[4] already take this approach by offering business training and mentoring to prospective and existing owners. As I discuss in Section 7, owners of basic businesses may gain more from help that builds on their specialized skills because these skills are more strongly associated with their business revenue and earnings.

2 Related Literature

Lazear's (2004, 2005) jack-of-all-trades (JOAT) hypothesis formalizes the idea that business owners benefit from being generalists, because they need to handle many tasks, such as marketing, operations, finance, and personnel management. Using data on Stanford MBA graduates, Lazear (2005) finds that individuals who have worked in more roles and studied more fields are indeed more likely to become business owners.

Subsequent to Lazear's (2004, 2005) work, many other researchers have tested the JOAT hypothesis in the context of entry into business ownership using a variety of measures of skill balance. Most studies proxy skill balance with counts of the employers, occupations, industries, or functional areas (departments such as marketing, finance, and R&D) a worker has experienced since starting her career (Wagner, 2003, 2006; Silva, 2007; Åstebro and Thompson, 2011; Stuetzer, Obschonka and Schmitt-Rodermund, 2013; Lechmann and Schnabel,

[4]SCORE originally stood for Service Corps of Retired Executives. It is a nonprofit organization that provides mentoring to small business owners, with mentors now including both active and retired business professionals.

2014; Chen and Thompson, 2016; Tegtmeier, Kurczewska and Halberstadt, 2016; Sorgner and Fritsch, 2018), while others use the variation in scores across a set of skill-specific aptitude tests (Hartog, Van Praag and Van Der Sluis, 2010; Aldén, Hammarstedt and Neuman, 2017). Using these measures, existing studies generally find that individuals with more balanced skills are more likely to enter business ownership, and Stuetzer, Goethner and Cantner (2012) find that start-up projects whose founders have more balanced skills make more progress toward founding a business. However, Silva (2007) finds that the relationship between skill balance and business ownership is no longer statistically significant after controlling for individual fixed effects, suggesting that persistent individual characteristics may account for part of the observed association.

A smaller literature examines outcomes after entry into business ownership using the measures of skill balance described above, mostly earnings and survival. For earnings, several studies find that more balanced skills are associated with higher earnings among business owners, more so than among wage workers (Hartog, Van Praag and Van Der Sluis, 2010; Åstebro, Chen and Thompson, 2011; Bublitz and Noseleit, 2014). For survival, business owners with more balanced skills tend to stay in business longer (Aldén, Hammarstedt and Neuman, 2017; Kurczewska and Mackiewicz, 2020). Business revenue has received little attention. An exception is Aldén, Hammarstedt and Neuman (2017), who examine turnover (sales revenue) and profit along with earnings and survival, and find that all are higher among business owners with more balanced abilities. Among studies that examine both entry and success, some find that the same measure relates differently to the two. Balanced abilities are associated with higher earnings for business owners but do not predict entry (Hartog, Van Praag and Van Der Sluis, 2010), whereas frequent job changes and more balanced work experience predict entry into business ownership but not higher income for business owners (Evans and Leighton, 1989; Åstebro and Thompson, 2011). Åstebro and Thompson (2011) attribute this pattern to a taste for variety rather than to balanced skills. I examine business ownership entry and success, measured by both revenue and earnings, within the same framework. Studies of both entry and success share three features that I revisit: all skills are treated as equally relevant to business ownership, all types of business ownership are pooled, and skill balance is proxied with measures that may capture other factors.

A related literature suggests that the types of skills workers accumulate matter for business ownership, beyond the balance of skills. Experience in running a business, such as start-up, management, and commercial experience, is associated with larger, faster-growing, and more profitable businesses (Unger et al., 2011; Ganotakis, 2012). Expertise relevant to the business also matters. Businesses perform better when the owner has relevant prior knowledge, gained in a related occupation (Koster and Andersson, 2018) or through early

exposure to the industry (Hvide and Oyer, 2025). Within the JOAT literature, Lechmann and Schnabel (2014) find that the work of business owners requires more expert-level skills, not only more basic-level skills, than that of wage workers, and some studies count functional areas that overlap partly with business skills, such as marketing and finance, but treat these counts as measures of skill balance (Stuetzer, Obschonka and Schmitt-Rodermund, 2013; Tegtmeier, Kurczewska and Halberstadt, 2016). Chen and Thompson (2016) go further and account for specific types of experience, such as finance, business administration, and senior management, finding that they largely explain the positive relationship between skill balance and founding a business. However, they measure each type of experience by whether a worker has ever held a job in that area, not by how much of it she has accumulated. I build on these findings by asking which types of skills matter for business ownership and why. I distinguish business skills from specialized skills and develop a framework that shows how each relates to entry and success. Following studies that combine workers' job histories with the skill requirements of occupations (Gathmann and Schönberg, 2010; Lise and Postel-Vinay, 2020), I measure how much of each type of skill a worker has accumulated across all her jobs.

Business ownership covers very different kinds of businesses, and the skills that matter for entry and success are likely to differ across them. Hurst and Pugsley (2011) show that most businesses are small and have no plans to grow or innovate, arguing that pooling all businesses can be inappropriate for testing standard theories of business ownership. In the NLSY79, Levine and Rubinstein (2017) show that incorporated business owners perform more nonroutine cognitive tasks and earn more than wage workers, whereas unincorporated business owners perform more manual tasks and earn less. Levine and Rubinstein (2018) also show that incorporated businesses start with substantially more capital than unincorporated businesses. Moreover, much of the self-employment recorded in the NLSY79 consists of contract or home-based work that respondents do not report as a business that they own (Light and Munk, 2018), and workers with relatively weak labor market attachment are more likely than other workers to run short-lived, low-capital, unincorporated businesses (Humphries, 2017). This evidence suggests that the skills relevant for entry into and success in business ownership may depend on the type of business, but the JOAT literature has largely not distinguished among these types.[5] I address this gap by distinguishing between two types of business ownership, basic and advanced, which I define on the basis of both incorporation status and start-up capital.

Finally, the occupation and employer counts commonly used in the JOAT literature

[5]Lechmann and Schnabel (2014) are an exception: they distinguish between business owners with and without employees and find that the positive relationship between the number of occupations and business ownership holds only among owners without employees.

measure job mobility, which is interpreted as a proxy for skill balance on the assumption that workers who move across occupations or employers accumulate more balanced skills. However, job mobility does not always reflect skill accumulation. It may reflect productive job shopping (Neal, 1999), but it may also reflect poor performance (Groes, Kircher and Manovskii, 2015). This second kind of mobility, which I refer to as job churning, is itself associated with entry into business ownership. In particular, workers who change jobs frequently or experience unemployment are more likely to become business owners (Evans and Leighton, 1989; Åstebro, Chen and Thompson, 2011), and those with poor labor market prospects tend to start small, short-lived businesses (Poschke, 2013). As a result, counts of occupations or employers may predict entry into business ownership even if skill balance plays no role. Studies that use the dispersion of ability test scores are not affected by job churning, but they examine a different concept, the balance of abilities measured before labor market entry (Hartog, Van Praag and Van Der Sluis, 2010; Aldén, Hammarstedt and Neuman, 2017). I instead focus on skills accumulated through work, comparing a direct measure of skill balance with the occupation count and using involuntary job losses to assess whether the count captures job churning.

3 Conceptual Framework

In this section, I present a static choice model to formalize how a worker's business skills and specialized skills (my two key regressors) affect her choice among wage employment, basic business ownership, and advanced business ownership, and how these skills affect her payoffs conditional on entry. The model provides a conceptual framework for understanding how business skills and specialized skills may differently predict entry into each type of business ownership and subsequent business revenue and earnings. I begin with a comparison of my model to Lazear's (2004, 2005) JOAT model, which forms the foundation for my approach.

3.1 Model Setup and Comparison to Lazear's Jack-of-All-Trades Model

In Lazear (2004, 2005), a worker has two arbitrary skills, x_1 and x_2, and chooses between working as a specialist in wage employment and becoming a business owner. A specialist is paid for her stronger skill, so her payoff is $\max(x_1, x_2)$. A business owner must perform many tasks herself, so her payoff is limited by her weaker skill; that is, her payoff is $\lambda \min(x_1, x_2)$, where λ is the market value of a business owner's talent. She becomes a business owner when $\lambda \min(x_1, x_2) > \max(x_1, x_2)$, which is most likely to hold when x_1 and x_2 are at similar levels. As a result, workers with balanced skills become business owners, and workers with one strong skill remain specialists in wage employment.

My framework keeps some of the basic structure of Lazear's model, in which a worker compares the payoff of wage employment with the payoff of business ownership. I follow Lazear in assuming that skills are given, but the two skills are not arbitrary. They are business skills, BX, and specialized skills, SX, the two types of skills that running a business likely requires, and the worker accumulates them through prior work experience.[6] Unlike Lazear, I distinguish two types of business ownership, which pay different returns to each skill. Like Lazear, I abstract from nonemployment, so the worker chooses one of three employment options, $j \in \{W, B, A\}$: wage employment (W), basic business ownership (B), or advanced business ownership (A).[7]

Each option pays a return on the worker's skills. I interpret the payoff y_j as the economic return the worker receives from option j. In wage employment, this is her earnings, and in business ownership, it is the income her business generates, whether she takes it home or reinvests it in the business. In Lazear's model, specialists are paid only for their stronger skill and business owners only for their weaker skill, which makes the balance between skills central to the choice. In my framework, each skill contributes to the payoff in every option, but at a rate that differs across options. Let r_j^{BX} and r_j^{SX} denote the returns to BX and SX in option j. I normalize both returns in wage employment to one, so that the returns in business ownership are measured relative to wage employment. The payoffs are additive

[6]Because the goal of the framework is to show how given skills affect the choice, I do not model how workers accumulate them. Relaxing this assumption is an area for future research. For example, an occupational choice or lifetime skill investment model could be extended to include business ownership as an option, so that workers decide how much of each skill to accumulate while knowing that they may later start a business.

[7]In the data, the alternative to business ownership pools wage employment and nonemployment (Section 4.2). Most entries come directly from wage employment. In the entry sample, 56 percent of entries into basic business ownership and 68 percent of entries into advanced business ownership occur while the worker still holds a wage job or within one month after it ends. A short break before starting a business, for example to prepare it, is close to a direct move from wage employment, whereas a long period without work before entry is a different path into business ownership. This second path is more common before basic business ownership. More than a year without work precedes 21 percent of entries into basic business ownership and 15 percent of entries into advanced business ownership, which is consistent with the role of job churning in Section 6.1.

in the two skills as follows:[8]

$$y_W = BX + SX, \tag{1}$$
$$y_B = r_B^{BX}\, BX + r_B^{SX}\, SX, \tag{2}$$
$$y_A = r_A^{BX}\, BX + r_A^{SX}\, SX. \tag{3}$$

Skills are not the only factors that determine whether a worker becomes a business owner. Workers also differ in wealth, risk tolerance, personality traits, preference for autonomy, business aspirations, and many other characteristics that affect the relative value of business ownership, independent of skill levels. I summarize all of these factors in two parameters, τ_B and τ_A, and I refer to them as *tastes*. Tastes allow workers with the same skills to make different choices, whereas in Lazear's model, skills alone determine who becomes a business owner. The taste for wage employment is normalized to $\tau_W = 0$. Tastes vary across workers with joint density $g(\tau_B, \tau_A)$, and τ_B and τ_A can be correlated, because, for example, a worker who values autonomy is likely to value both types of business ownership. Tastes change how attractive an option is, but they do not change the returns to skills in that option. Because I take both skills and tastes as given and model their effect on the choice among W, B, and A, tastes do not influence how much of each skill a worker has in this framework.[9]

The worker chooses the option with the highest utility, where the utility of option j is its payoff plus her taste for it, $U_j = y_j + \tau_j$.[10] The utility gain from choosing option $j \in \{B, A\}$ over wage employment is $U_j - U_W = \tau_j - c_j$, where c_j denotes the opportunity cost of option j. Using equations (1)–(3), this opportunity cost can be expressed as

$$c_j = y_W - y_j = (1 - r_j^{BX})\, BX + (1 - r_j^{SX})\, SX, \qquad j \in \{B, A\}. \tag{4}$$

The worker prefers option j to wage employment when $\tau_j > c_j$.

Figure 1 shows the worker's choice for each possible pair of tastes (τ_B, τ_A). She chooses wage employment when neither taste exceeds its opportunity cost; that is, when $\tau_B < c_B$ and $\tau_A < c_A$. She chooses basic business ownership when $\tau_B > c_B$ *and* the gain from basic business ownership exceeds the gain from advanced business ownership, $\tau_B - c_B > \tau_A - c_A$,

[8]The predictions in Section 3.2 do not require additivity. They depend only on how the marginal return to each skill compares across options, for example, whether advanced business ownership pays more for BX than wage employment does. Additivity means that the return to each skill does not depend on how much of the other skill the worker has, so these comparisons are the same for every worker. If the two skills are complements instead, the marginal return to each skill depends on the level of the other skill, and the predictions hold as long as the comparisons hold at the worker's skill levels.

[9]In reality, tastes may affect how much of each skill a worker accumulates. For example, a worker with a strong preference for autonomy may deliberately choose a career in which she can accumulate more business skills, in anticipation of becoming a business owner. I discuss this in Section 5.3.

[10]This form captures the trade-off between earnings and nonpecuniary benefits, such as being one's own boss. Studies of business ownership find that such benefits help explain why many workers become business owners despite lower earnings (Hamilton, 2000; Hurst and Pugsley, 2011).

and she chooses advanced business ownership in the opposite case. The three regions meet at the point (c_B, c_A). Along the diagonal boundary, $\tau_B - c_B = \tau_A - c_A$ and the worker is indifferent between basic and advanced business ownership. Although the worker's choice is determined by her tastes (relative to opportunity costs), her skills determine the opportunity costs (c_B, c_A) and, in turn, the boundaries shown in Figure 1.

3.2 Empirical Predictions

I now use the framework to show how BX and SX are expected to affect the choice among wage employment, basic business ownership, and advanced business ownership, as well as payoffs after entry. In the model, a worker's choice is known once her skills and tastes are given. In the data, I observe her skills but not her tastes, so I treat her tastes as a draw from the density g and predict the probability of each choice. The probability that she chooses option j, P_j, is the probability that her tastes lie in the region of Figure 1 that corresponds to option j.[11] Because skills enter the choice only through the opportunity costs, the effect of a skill on P_j works in two steps. The skill changes the opportunity costs (c_B, c_A), and the change in the opportunity costs then changes the probabilities.

In the first step, an increase in skill $k \in \{BX, SX\}$ changes the opportunity cost of option j by $\partial c_j / \partial k = 1 - r_j^k$, from equation (4). That is, skill k makes option j more attractive relative to wage employment when its return is higher in option j than in wage employment ($r_j^k > 1$). In the second step, a change in the opportunity costs shifts the boundaries in Figure 1, which changes the probabilities. Consider an increase in c_B. The vertical boundary between wage employment and basic business ownership and the diagonal boundary both shift to the right. If her tastes lie just to the right of the vertical boundary, she now chooses wage employment, and if they lie just below the diagonal boundary, she now chooses advanced business ownership. As a result, the probability that she chooses basic business ownership falls, and the probability that she chooses advanced business ownership rises. An increase in c_A works the same way in the other direction. Hence

$$\frac{\partial P_B}{\partial c_B} < 0, \qquad \frac{\partial P_A}{\partial c_B} > 0, \qquad \frac{\partial P_B}{\partial c_A} > 0, \qquad \frac{\partial P_A}{\partial c_A} < 0. \tag{5}$$

In both cases, the wage employment region grows, so an increase in either opportunity cost lowers the probability that she becomes a business owner. These signs hold for any taste density g, because an increase in c_B always shrinks the region of basic business ownership and enlarges the regions of wage employment and advanced business ownership. By contrast,

[11] If each taste parameter is the sum of a constant and an independent, individual-specific type I extreme value term, the framework becomes the multinomial logit model of entry that I estimate in Section 5.1. The coefficient on skill $k \in \{BX, SX\}$ in the log odds of option j relative to wage employment is then $r_j^k - 1$.

the signs of the effects of skills, BX and SX, on entry into and success in business ownership depend on the sizes of the returns. Below, I discuss the possible cases, argue which is most plausible, and state the prediction that follows.

Business skills and entry into business ownership. Combining equations (4) and (5), the effect of BX on the probability of entering each type of business ownership is

$$\frac{\partial P_A}{\partial BX} = \frac{\partial P_A}{\partial c_A}(1 - r_A^{BX}) + \frac{\partial P_A}{\partial c_B}(1 - r_B^{BX}), \tag{6}$$

$$\frac{\partial P_B}{\partial BX} = \frac{\partial P_B}{\partial c_B}(1 - r_B^{BX}) + \frac{\partial P_B}{\partial c_A}(1 - r_A^{BX}). \tag{7}$$

If advanced business ownership pays more for BX than wage employment does ($r_A^{BX} > 1$), and basic business ownership pays less ($r_B^{BX} < 1$), both terms in equation (6) are positive and both terms in equation (7) are negative. I argue that this is the most plausible case. An owner of an advanced business uses her business skills to coordinate workers, allocate resources, and manage the business as a whole, allowing these skills to generate returns across the entire business (Lucas, 1978). By contrast, a wage worker applies her business skills within the scope of her assigned responsibilities, limiting the range of activities over which these skills generate returns. Basic business ownership involves simpler operations and less need for coordination and management, so the returns to business skills are likely to be smaller. Other cases are also possible. If large firms offer managers higher returns to business skills than advanced business ownership does, then $r_A^{BX} < 1$, and BX increases the opportunity cost of both types of business ownership. However, as long as advanced business ownership pays more for BX than basic business ownership does ($r_A^{BX} > r_B^{BX}$), BX still shifts the diagonal boundary toward basic business ownership, making advanced business ownership more likely relative to basic.

> **Prediction 1.** Business skills raise the probability of entering advanced business ownership and lower the probability of entering basic business ownership.

Specialized skills and entry into business ownership. An increase in SX changes the opportunity costs of basic and advanced business ownership by $1 - r_B^{SX}$ and $1 - r_A^{SX}$, respectively. If both types of business ownership pay less for SX than wage employment does ($r_B^{SX} < 1$ and $r_A^{SX} < 1$), SX increases both opportunity costs and lowers the probability of becoming a business owner. I argue that this is the most plausible case. A business owner's expertise typically applies to only a subset of the tasks required to run the business, whereas a wage worker can devote more of her time to tasks that use her specialized skills. Other cases are possible as well. When a business relies directly on its owner's expertise, as in a professional practice or skilled trade, business ownership may offer higher returns to

specialized skills than wage employment does. Even in the most plausible case, the framework does not determine the effect on each type separately. A rise in c_B leads some workers who would have chosen basic business ownership to choose advanced business ownership instead, and a rise in c_A does the reverse. Because the wage employment region also grows, a rise in c_B can raise the probability of entering advanced business ownership even as it lowers the probability of entering business ownership of either type.

> **Prediction 2.** Specialized skills lower the probability of entering business ownership of either type.

Skills and success in business ownership. Once the worker owns a business, the effect of a skill on her payoff is simply the return to that skill, r_j^{BX} or r_j^{SX} (equations (2) and (3)). Both BX and SX raise payoffs in both types of business ownership as long as their returns are positive, which is plausible in all cases discussed above. For the reasons given for Prediction 1, business skills are likely to raise payoffs more in advanced than in basic business ownership ($r_A^{BX} > r_B^{BX}$). This comparison also shows the main difference between entry and success. Entry depends on how a return compares with wage employment ($r_j^k - 1$), whereas success depends on the return itself (r_j^k). As a result, a skill with $0 < r_j^k < 1$ can lower entry into business ownership and still improve success once the worker owns a business. This cannot happen in Lazear's model, where the skill that raises a business owner's payoff, her weaker skill, also makes entry more likely.

> **Prediction 3.** Business skills and specialized skills both raise payoffs in business ownership, and business skills raise payoffs more in advanced than in basic business ownership.

4 Data

4.1 Data Sources

In conducting my empirical analysis, I combine data from the 1979 National Longitudinal Survey of Youth (NLSY79) with data from O*NET (described below). The NLSY79 began in 1979 with a nationally representative sample of 12,686 individuals born between 1957 and 1964 and residing in the United States. The respondents were surveyed annually from 1979 to 1994 and biennially thereafter. I use survey waves from 1979 through 2020.

The NLSY79 is uniquely well-suited for my study for a number of reasons. First, it follows respondents over a long horizon. I am able to track respondents' work histories from school exit until their last interview date, which in many cases affords a four-decade observation window. This long coverage is particularly valuable for studying business ownership, which

is often entered in the mid or late career. Second, the survey tracks employment, occupation, and nonemployment spells in detail. This enables me to construct precise measures of the experience and skills accumulated at any point in time and to observe transitions into and out of each job, where a job is a spell with one employer. Third, the survey identifies which jobs are self-employment or self-owned businesses. In addition to the traditional class-of-worker variable, which classifies each job as self-employment or wage employment, a separate business ownership module which was introduced in 2010 identifies all businesses that respondents own and links them to specific job spells. Fourth, it provides ample detail on jobs (such as earnings and hours worked), businesses (such as revenue, incorporation status, and start-up capital), and individuals (such as educational attainment, cognitive and noncognitive abilities, family and household structure, and wealth).

I also use data from version 18.1 of the Occupational Information Network (O*NET), which is administered by the U.S. Department of Labor.[12] The O*NET database contains a wealth of occupation-specific information. I use the so-called "level" and "importance" scores, which indicate both the amount of and importance of dozens of skill and knowledge categories in each of 873 occupations. By linking these occupation-specific scores to each NLSY79 sample member's occupations, I am able to construct my time-varying measures of business skills, specialized skills, and skill balance.

4.2 Sample Construction

As discussed in detail in Section 5, I estimate three models: a competing-risks model for the decision to transition into business ownership, and both a business revenue model and an earnings model for workers who are business owners. I construct a separate sample of NLSY79 respondents for each model.

Sample for the entry model. To model the year-by-year decision of individuals who are not business owners (that is, wage workers and the nonemployed) to enter either basic or advanced business ownership, I define the career start date for each respondent as the month and year in which the first nonenrollment spell lasting at least 12 months begins. Starting with the full sample of 12,686 NLSY79 respondents, I first exclude 392 individuals who leave the survey before their career start date or before holding at least one job after the start date. Second, I exclude 936 individuals with incomplete work histories after the career start date (for example, missing job start or stop dates). Third, I exclude 4,037 individuals who exit the survey before 2010, when the business ownership module was added. Fourth, I exclude 25

[12]This information is obtained on an ongoing basis, starting in 1998, via surveys of randomly sampled workers and from input provided by occupational experts. As a result, the data represent occupational conditions over the last three decades only, and are not year-specific.

individuals who are only business owners (without any wage employment) throughout their observed careers. Finally, I exclude 180 individuals with left-censored careers, meaning they started their careers before the survey began tracking work experiences. These restrictions leave a sample of 7,116 individuals.

Each of these 7,116 sample members contributes one person-year observation to the sample for every 12-month period that they are not business owners. This sequence of observations begins at the career start date, and ends when they enter business ownership or are last interviewed. However, individuals who leave business ownership reenter the sample and contribute additional observations. The final panel contains 251,214 person-year observations for 7,116 individuals.

Samples for the business revenue and earnings models. To model business revenue, I return to the sample of 7,116 individuals described above and restrict it to those who ever experience business ownership. Each of these individuals contributes one observation for each business and year for which business revenue is reported. For businesses without a yearly report, such as those that ended before 2010, I use the revenue that respondents report retrospectively for a typical year. A limitation of NLSY79 data is that business revenue is only collected from 2010 onward, and only for jobs identified by respondents as self-owned businesses.[13] Because of this limitation, the resulting sample contains only 2,430 observations for 938 individuals with 1,221 basic businesses, and 2,465 observations for 832 individuals with 1,027 advanced businesses. As discussed below, I use the more broadly available earnings data to assess how these sample restrictions affect the results.

To model earnings among individuals who are currently business owners (and/or self-employed, per footnote 13), I use a sample selection strategy similar to the one I use for the revenue sample. Starting with the same individuals, I allow each individual to contribute one observation for each business and year for which weekly earnings are available. This produces a sample of 10,383 observations for 2,370 individuals with 4,118 basic businesses, and 5,938 observations for 1,103 individuals with 1,554 advanced businesses. I also estimate the earnings model on the subset of these observations for which revenue is also available.

[13]From 1979 onward, respondents report the class of worker for every job held, which identifies a subset of jobs as self-employment. Starting in 2010, respondents retrospectively report every business owned since age 18, and they report new businesses in each later interview. Many jobs are classified both as self-employment and as a business, but many are classified as only one of the two (Light and Munk, 2018). Owners of incorporated businesses may report themselves as employees of their own firms, while workers in small, home-based businesses often do not report their work as a business. I use the universe of self-employment jobs and self-owned businesses to model entry into business ownership and, as described below, to model earnings, but I can only use jobs classified as businesses (many of which are also classified as self-employment) in the revenue analysis. I exclude the few businesses that cannot be linked to any job in the employer history.

4.3 Business Types and Outcome Variables

In modeling entry into business ownership, business revenue, and earnings among business owners, I distinguish between two types of business ownership (basic and advanced) using a combination of incorporation status and start-up capital at entry.[14] Many studies separate business owners by incorporation status (Levine and Rubinstein, 2017, 2018; Humphries, 2017; Can, 2022; Can and Fossen, 2022). Incorporation status is an important dimension because it captures the owner's legal commitment to the business. However, the decision to incorporate can also reflect tax or liability considerations, and businesses with the same legal form can differ widely in how much they invest at entry. On average, incorporated businesses start with more capital. Among businesses with positive start-up capital, the median is \$12,427 and the mean is \$111,309 for incorporated businesses, compared with \$3,883 and \$27,143 for unincorporated businesses. Nevertheless, as Figure 2 shows, many incorporated businesses start with little capital, while some unincorporated businesses start with \$50,000 or more. Start-up capital captures a dimension of business ownership that incorporation status alone does not: the financial resources committed at entry. I use both measures to classify businesses.

I classify a business as basic if it is unincorporated *and* its start-up capital is less than \$5,000 in 2010 dollars.[15] I classify a business as advanced if it is incorporated *or* its start-up capital is at least \$5,000. The cutoff of \$5,000 is approximately the median value among all businesses with positive start-up capital.

Unsurprisingly, basic business ownership is far more common than advanced business ownership. Of the 4,566 businesses started in the entry sample, 76 percent are basic and 24 percent are advanced. As indicated in Section 4.2, basic businesses account for 73 percent of the businesses used to model earnings, but only 54 percent of those used to model business revenue, given the restrictions on that sample.

When modeling entry into business ownership, I use the classification just described to determine whether, on a year-by-year basis, a worker who is not currently a business owner remains in her current state, becomes a basic business owner, or becomes an advanced business owner. Because I define entry as a transition from not owning a business to owning one, I do not count a direct move from one business to another as an entry.

When modeling outcomes among business owners, I use both business revenue and earnings. As noted in Section 4.2, business revenue is collected only from 2010 onward and only

[14] Another common approach to classifying business ownership is to consider firm size, typically measured by the number of employees. Firm size was not collected from 1981 to 1985 in the NLSY79, which may lead to inconsistent measurement across career stages.

[15] Throughout this analysis, I use the annual Consumer Price Index for four geographic regions (Northeast, Midwest, South, and West) to express all dollar amounts in 2010 dollars.

for jobs that respondents identify as businesses, while earnings are reported for every job from 1979 onward. At each interview, respondents report their current or most recent "rate of pay" for every job held since the prior interview. Some report this as weekly earnings. In instances where pay is reported as hourly, biweekly, monthly, or in other units, I use reported weekly hours to construct a measure of weekly earnings. I use weekly rather than hourly earnings because business owners can adjust their hours more flexibly, so differences in earnings may reflect both hourly returns and hours worked. Both outcomes enter the models in logs. Because some business owners report zero weekly earnings, I add one to each outcome before taking the log.

4.4 Independent Variables

I first discuss the three O*NET-based skill measures—business skills, specialized skills, and skill balance—which are the key independent variables. All three are time-varying. For each worker, I compute them in every year from the career start date onward, using her occupational history up to that year. The entry model uses their values in each year in which the worker is at risk of entry, and the revenue and earnings models use their values at the start of each business. I then explain how I construct the occupation count, a conventional measure of skill balance, and the baseline regressors included in each model.

4.4.1 Business Skills

To construct my measure of business skills, BX_{it}, I combine information on the business-skill intensity of occupations with workers' occupational histories. I proceed in three steps. First, I identify the O*NET skill and knowledge elements that capture business skills. Second, I use these elements to measure the business-skill intensity of each occupation. Third, I combine these intensities with the experience worker i has accumulated in each occupation from the career start date to time t, which gives BX_{it}.

In step 1, I identify those O*NET skill and knowledge elements that capture three components of establishing and operating a business: core business knowledge, managerial skills, and interpersonal skills. Core business knowledge includes economics and accounting, sales and marketing, production and processing, customer and personal service, and service orientation. Managerial skills consist of administration and management, personnel and human resources, and the management of financial, material, and personnel resources. Interpersonal skills consist of social perceptiveness, coordination, persuasion, negotiation, and instructing; the first four are the items that Deming (2017) uses to measure social skills. In making this selection, I use the O*NET elements that rank highest for chief executives (CEOs) as a reference, because the CEO occupation encompasses key activities of business ownership,

including making financial and strategic decisions, managing staff and resources, and interacting with customers or clients. This process identifies 15 of O*NET's 68 total skill and knowledge elements as what I refer to as "business elements." Table A1 lists all 68 elements and identifies the 15 selected to construct the business skill measure.

In step 2 of the construction process, I use these 15 business elements to assign each occupation a business-skill intensity. O*NET rates each of its 68 skill and knowledge elements in each of its 873 occupations on two dimensions: a level score from 0 to 7, measuring how much of the element the occupation requires, and an importance score from 1 to 5, measuring how central the element is to the occupation. I rescale both scores to range from 0 to 1 and refer to their product as the element score.[16] I use the element scores of the 587 occupations held by NLSY79 respondents.[17] Specifically, I calculate the average element score of the 15 business elements for each occupation and normalize this average by the highest value across these occupations. This normalization sets the most business-intensive occupation (CEO) to 1 and expresses the intensity of every other occupation relative to that benchmark.

In step 3, I combine these occupational intensities with workers' occupational histories to construct BX_{it}. For worker i at time t, I define

$$BX_{it} = \sum_{o \in O_i} \omega_o^{BX} \cdot OccExp_{iot}, \tag{8}$$

where O_i is the set of occupations worker i has ever held, and $OccExp_{iot}$ is her cumulative experience in occupation o from the career start date to t, measured in years. The weight ω_o^{BX} is the normalized business-skill intensity of occupation o.

For example, ten years of uninterrupted employment as a CEO, the most business-intensive occupation with an intensity of 1, yield a BX_{it} of 10. Ten years as a purchasing manager, whose intensity is 0.81, give a BX_{it} of 8.1, while ten years as an agricultural product grader or sorter, one of the least business-intensive occupations with an intensity of 0.08, give a BX_{it} of 0.8. These examples illustrate how BX_{it} assigns greater weight to experience acquired in occupations with higher business-skill intensity. Panel A of Table 1 reports summary statistics of BX_{it}.

A potential concern is that BX_{it} captures general skill requirements rather than business-specific skills, because occupations that require and use more business skills are likely to also

[16]The element score takes into account both how much of an element the occupation requires and how central the element is to the work. In practice, the two scores are highly correlated. Across the occupations held by workers in my sample, the correlation between level and importance is 0.96, and each has a correlation of 0.94 with their product.

[17]The NLSY79 codes occupations with the 1970 Census classification through 2000 and the 2000 Census classification thereafter, whereas O*NET uses the 2010 Standard Occupational Classification (SOC). I link both NLSY79 classifications to the 2010 SOC using the crosswalks developed by Lise and Postel-Vinay (2020).

require more education, training, and prior experience. To assess this concern, I use O*NET job zones, a five-level classification of the preparation an occupation requires, ranging from Zone 1 (little or no preparation) to Zone 5 (extensive preparation). I add to the entry model each worker's years of experience in Zone 4 and 5 occupations, most of which require a bachelor's degree or more. Table A4 reports the results. On its own, experience in these occupations is associated with lower entry into basic business ownership and higher entry into advanced business ownership. Once BX_{it} is included, these estimated effects decline substantially toward zero, while the estimated effects of BX_{it} remain largely unchanged. This suggests that BX_{it} captures skills specific to establishing and operating a business rather than general skill requirements.

4.4.2 Specialized Skills

The specialized skills variable, SX_{it}, measures the extent to which a worker's experience up to time t matches the specific expertise required by her target business. For example, a worker who starts a law firm needs expertise in law and government, whereas a worker who starts a plumbing business needs expertise in mechanical work and building/construction. Because the expertise required varies across businesses, the measure is tailored to each worker's target occupation. I define the *target occupation* as the occupation held at the start of the business for workers who enter business ownership, and as the occupation held 18 years into the career for those who never do, which corresponds to the average point of entry into business ownership in the sample.[18]

To determine which specialized skills are required in each target occupation, I rank all 68 O*NET skill and knowledge elements (see Table A1) by their element scores, defined in Section 4.4.1 as the product of the rescaled importance and level scores. I select the top one to three elements, using natural breaks in the ranking to identify elements that stand out from the rest. I then exclude elements that rank near the top across most occupations, because broadly required skills do not distinguish a worker's area of specialization. For example, the top three elements for lawyers are law and government, English language, and speaking. Because English language and speaking rank near the top in many occupations, I keep only law and government as the specialization element. For plumbers, the top two elements are mechanical and building/construction, and both are kept.[19]

[18] Alternatively, I could define the target occupation for workers who do not enter business ownership as the occupation in which they have spent the most time. The estimates are robust to which measure I use.

[19] This procedure is similar in spirit to TF-IDF (term frequency–inverse document frequency) weighting in natural language processing, where element scores capture the strength of an element within an occupation, while excluding elements that rank near the top across most occupations removes elements that are not distinctive to a particular occupation.

I construct specialized skills using the same approach as the business-skills measure, but with weights tailored to each worker's target occupation. For worker i at time t, I define

$$SX_{it} = \sum_{o \in O_i} \omega_{io}^{SX} \cdot OccExp_{iot}, \tag{9}$$

where the weight ω_{io}^{SX} is the average, across worker i's specialization elements, of occupation o's element score divided by that element's highest score among the occupations held by sample members. An occupation receives a weight of 1 if no other occupation uses any of worker i's specialization elements more intensively. Because the weights depend on the target occupation, two workers with identical work histories can have different values of SX_{it} if their target occupations differ. As a result, the measure captures experience in occupations that develop the specific expertise required by a worker's target business, rather than work experience in general. See Panel A of Table 1 for summary statistics of SX_{it}.

4.4.3 Skill Balance

My direct measure of skill balance, which I call the *balanced skill count* (BSC_{it}), captures how evenly a worker's accumulated skills are distributed across O*NET's 68 skill and knowledge elements. I construct it in three steps. First, I divide each element score by its highest value across the occupations held by workers in my sample, so that all elements are measured on the same scale. Second, for each element, I compute the worker's accumulated skill by summing these rescaled scores across all occupations she has held, weighted by the years she spent in each. Third, I keep the 30 elements in which she has accumulated the most skill.[20] Letting p_{its} denote the share of element s in her accumulated skill across these 30 elements, I summarize the distribution using the exponentiated Shannon index

$$BSC_{it} = \exp\left(-\sum_{s} p_{its} \log p_{its}\right). \tag{10}$$

The variable BSC_{it} ranges from 1, when all accumulated skill is concentrated in a single element, to 30, when it is distributed equally across all 30 elements. A higher value indicates a more balanced skill portfolio, while a lower value indicates greater concentration in a few elements.[21]

[20] I exclude the remaining 38 elements because workers typically accumulate little or no skill in some of them. Including all 68 would mechanically make every worker's skills appear unbalanced, even when they are evenly distributed across the elements relevant to her work experience. On average, the top 30 elements account for about 80 percent of a worker's accumulated skill across all 68 elements.

[21] A common alternative is the inverse Herfindahl–Hirschman index (HHI), $1/\sum_s p_{its}^2$. Because the HHI squares skill shares, it is more sensitive to changes in larger shares and less sensitive to changes in smaller shares. In contrast, the exponentiated Shannon index places relatively more weight on smaller shares, making it better suited to Lazear's model, in which a business owner's payoff depends on her weakest skill.

To illustrate how my direct measure of skill balance (BSC_{it}) compares to the conventional occupation count (discussed in Section 4.4.4), consider two workers, each with 15 years of experience divided equally across three occupations. The first worker has worked as a bookkeeping clerk, an accountant, and a financial manager. Because these occupations draw heavily on overlapping elements, such as mathematics and economics/accounting, her accumulated skill portfolio is concentrated in a relatively small set of elements. The second worker has worked as an accountant, a registered nurse, and a construction supervisor. These occupations draw on more distinct elements, spanning economics/accounting, medicine/dentistry, and building/construction. As a result, her accumulated skill portfolio is more evenly distributed across elements, resulting in a higher BSC_{it}, even though both workers have an occupation count of three. See Panel A of Table 1 for summary statistics of BSC_{it}.

4.4.4 Number of Occupations and Involuntary Job Losses

To compare my direct measure of skill balance with the approach used in prior work, I also construct the occupation count, one of the most common proxies for skill balance. The occupation count is the number of distinct occupations a worker has held from the career start date to time t. I count occupations using the 23 two-digit occupations in the 2010 Standard Occupational Classification (SOC). The count increases when a worker enters a two-digit occupation she has not previously worked in, and returning to a previously held occupation does not increase the count. Unlike business skills, specialized skills, and the balanced skill count, this measure does not use O*NET data.

I also construct a time-varying count of involuntary job losses to assess whether the occupation count captures job churning rather than skill balance. For each job that ends, the NLSY79 records the reason for separation. I classify a job ending as an involuntary job loss if the worker was laid off, discharged, or fired, if the plant closed, or if a temporary job, seasonal job, or program ended. The count is the number of involuntary job losses from the career start date to time t. See Panel A of Table 1 for summary statistics of the occupation count and the number of involuntary job losses.

4.4.5 Baseline Regressors

In my empirical analysis, I include a rich set of control variables. The choice of controls follows a large literature on the determinants of business ownership, including risk tolerance (Ahn, 2010; Hsieh, Parker and van Praag, 2017), access to credit and household wealth (Hurst and Lusardi, 2004; Fairlie and Krashinsky, 2012), and family background (Hout and Rosen, 2000; Hvide and Oyer, 2025).

I control for past work experience, measured in two ways. Potential experience is the time elapsed since the career start date, and actual experience is the time actually worked, adjusted for the average weekly hours worked on each job. Both are measured in years and enter with their squares. I also control for the number of past business ownership spells.

Among individual characteristics, I control for gender, race/ethnicity (Hispanic, Black, or neither), marital status (never married, currently married, or previously married), whether the respondent is cohabiting with an unmarried partner, and whether the youngest child in the household is age six or younger.[22] I also control for education (less than high school, high school graduate, some college, or college or above) and the Armed Forces Qualification Test (AFQT) percentile as a measure of cognitive ability. As measures of noncognitive traits, I include risk tolerance (four categories based on the NLSY79's income-gamble questions), sociability at age six (four categories from extremely shy to extremely outgoing), Rotter's locus of control score, and two of the Big Five personality traits, extraversion and openness to experience, measured with self-rated items from the Ten-Item Personality Inventory (TIPI). I also use whether the respondent considers herself an entrepreneur as a proxy for her taste for business ownership. Because this measure is available only from 2010 onward and may reflect prior business ownership experience, I use it only in the sensitivity analysis in Section 5.3.

For family background, I control for net family wealth, net family income, whether the respondent's father had at least some college education, whether the father worked in a professional or managerial occupation in 1979, and whether any family member owned a business. Because net family wealth is available only from 1985 onward, I impute it for earlier years.

In the revenue and earnings models, I include all of these controls except net family wealth and income, because both measures include business income for business owners and thus partly capture the outcomes of interest. I also control for tenure in the current business and its square, and for region and industry fixed effects. In these two models, experience is measured at the start of each business, like the skill measures.

The time-varying controls are experience, the number of past business ownership spells, education, marital and cohabitation status, the young-child indicator, net family wealth and income, risk tolerance, the Rotter score, region, industry, and tenure in the current business; all other controls are time-invariant. See Panels B–E of Table 1 for summary statistics of these baseline regressors.

[22]Having a young child can motivate entry into self-employment as a means to balance work and family, especially for women. See Lim (2019); Lim and Michelmore (2018).

4.5 Descriptive Statistics

Before turning to my empirical strategy and findings, I establish some basic patterns in the data. The descriptive statistics in this subsection (Tables 2–6) show (a) how many individuals become business owners and how many enter business ownership more than once; (b) which occupations are most common in each type of business ownership; (c) how entrants into basic and advanced business ownership differ from each other and from workers in years without entry; (d) how basic and advanced businesses differ in their characteristics; and (e) how strongly my skill and count measures are correlated with each other.

Table 2 shows how often the 7,116 individuals in the entry sample enter business ownership (Panel A) and which types of businesses they enter (Panel B). Over the sample period, 2,950 of these individuals (41.5 percent) enter business ownership at least once. This cumulative share reflects the fact that the sample follows individuals for nearly 40 years and includes business ownership in both main and side jobs. For comparison, in July 2010, when sample members are aged 45–53, 13.4 percent of employed individuals in the sample own a business as their main job, close to the 13.5 percent of workers aged 45–54 who are self-employed in the 2009 Current Population Survey (Hipple, 2010). Including side businesses raises the share in the sample to 17.0 percent. Among individuals who ever enter business ownership, 37.8 percent do so more than once. Basic businesses are also far more common at the individual level. Of all individuals in the sample, 28.0 percent enter only basic businesses, compared with 8.4 percent who enter only advanced businesses. Another 5.1 percent enter both types at different points in their careers, representing about one in eight individuals who ever enter business ownership. Together, these patterns show that many individuals enter business ownership at some point in their careers, and that more than a third of those who enter do so more than once. For this reason, my entry model counts every entry, not only the first, and allows the same individual to enter different types of business at different times.

Table 3 shows what kinds of businesses these individuals run by listing the eight most common occupations in each type of business. The lists show that business ownership in the sample is not limited to small personal-service work and that the two types differ in the kind of work involved. Basic businesses are concentrated in personal services and manual trades, such as childcare, landscaping, cleaning, painting, and construction labor. Advanced businesses are more concentrated in managerial occupations, such as managers, chief executives, retail sales supervisors, and food service managers. A few occupations, such as sales workers, carpenters, and managers, appear in both lists. These occupations are common among business owners of both types, so occupation alone does not determine the type of business.

Table 4 shows that basic entrants start their businesses earlier in their careers than advanced entrants. On average, basic entrants are 35.0 years old and have 10.9 years of actual experience when they enter, whereas advanced entrants are 38.5 years old and have 15.3 years of actual experience. Advanced entrants have also accumulated more business skills. Their average BX_{it} is 6.0, compared with 3.6 for basic entrants. Because BX_{it} grows with time in the labor market, part of this gap reflects the later entry of advanced entrants, but the gap remains when business skills are measured per year of actual experience. Advanced entrants have accumulated 0.38 per year, compared with 0.31 for basic entrants and 0.32 in person-years without entry, so their careers have been concentrated more heavily in business-intensive occupations. Specialized skills per year and the balanced skill count, in contrast, differ little between the two types. The two types also differ clearly in job churning. Basic entrants have worked in 0.71 occupations per year of actual experience and have had about twice as many involuntary job losses per year as advanced entrants, whereas advanced entrants have worked in 0.48 occupations per year, fewer than the 0.53 in person-years without entry. Finally, advanced entrants are more educated and score higher on the AFQT, with 31 percent holding a college degree and an average AFQT percentile of 51, compared with 18 percent and 43 for basic entrants.

In Table 5, I compare the characteristics of basic and advanced businesses. Advanced businesses generate more revenue, provide higher earnings to their owners, and last longer than basic businesses. Excluding top-coded values, average annual business revenue is \$215,000 for advanced businesses, more than three times the \$63,000 for basic, and revenue is top-coded for 9 percent of advanced businesses but for only 1 percent of basic ones. Owners of advanced businesses earn \$844 per week, almost twice the \$443 earned by owners of basic businesses, and work about nine more hours per week. Advanced businesses last 7.6 years on average, compared with 3.9 years for basic businesses, and 49 percent of them last five years or more, compared with 23 percent of basic businesses. At the last interview, 32 percent of advanced businesses are still ongoing, compared with 12 percent of basic ones.

Because my skill and count measures enter the models together, Table 6 reports how strongly they are correlated with one another. All five measures grow with time in the labor market, so the table reports partial correlations that control for actual experience. Panel A uses the person-years in the entry sample, and Panels B and C use basic and advanced businesses in the earnings sample, measured at business start. The patterns are similar across the three panels. The correlations between BX_{it} and SX_{it} are moderate, ranging from 0.40 to 0.56, partly because in some occupations, such as sales, the specialization elements are business elements. The two measures are therefore related but distinct. The correlations of BSC_{it} with BX_{it} are weak, between 0.15 and 0.21, and those with SX_{it} are close to

zero, so skill balance is separate from how much business or specialized skill a worker has accumulated. The occupation count follows a different pattern. Its correlations with BSC_{it} range from 0.18 to 0.25, but those with involuntary job losses are higher, between 0.36 and 0.48, and its correlations with BX_{it} and SX_{it} are negative. This pattern suggests that the occupation count, a common proxy for skill balance in tests of the JOAT hypothesis, partly reflects job churning rather than skill balance alone. Taken together, these correlations indicate that BX_{it}, SX_{it}, and BSC_{it} capture distinct aspects of work experience, each with enough independent variation to be included in the same model.

5 Empirical Strategy

5.1 Entry into Business Ownership

I use a sequential, discrete-time, competing-risks model for the probability of entry into business ownership. In each year, a worker who is not currently a business owner faces three options. She can enter basic business ownership, enter advanced business ownership, or stay out of business ownership. The latent value of option j for worker i in year t is

$$V_{ijt} = \alpha_{0j} + \alpha_{1j} BX_{it} + \alpha_{2j} BX_{it}^2 + \alpha_{3j} SX_{it} + \alpha_{4j} SX_{it}^2 + \alpha_{5j} BSC_{it} + \alpha_{6j} BSC_{it}^2 + X_{it}\alpha_{Xj} + \varepsilon_{ijt}, \tag{11}$$

where BX_{it} is business skills, SX_{it} is specialized skills, BSC_{it} is the balanced skill count, and X_{it} is the set of baseline regressors described in Section 4.4.5 (Panels B–D of Table 1). The worker chooses the option with the highest latent value. The coefficients depend on j because each skill measure and each covariate can affect entry into basic and advanced business ownership differently. Each skill measure enters with a quadratic term because the return to an additional year of a skill may change as the stock grows. Besides this main specification, I estimate versions in which BX_{it}, SX_{it}, and BSC_{it} enter individually and jointly, and versions that include the number of occupations held to date and the number of involuntary job losses (Section 4.4.4). I assume that the errors ε_{ijt} are independent across options and follow a type I extreme value distribution, which yields a multinomial logit (MNL) model. For comparison, I also estimate a binary logit that pools entry into basic and advanced business ownership. Standard errors are clustered at the individual level to allow for correlation in the errors over time for a given individual.

5.2 Business Revenue and Earnings Models

As noted earlier, I model both business revenue and earnings among business owners as two alternative measures of business success. Revenue measures the scale of the business,

but it does not subtract costs, and the data limitations described in Section 4.2 leave a restricted sample. Weekly earnings are available for nearly every business and often for several years, but they understate the owner's payoff when owners reinvest profits or pay themselves little for tax reasons. Because neither outcome measures the payoff y_j in the conceptual framework exactly, I examine both. I estimate both the revenue model and the earnings model separately for observations associated with basic and advanced businesses, because the conceptual framework (Section 3) predicts that business skills and specialized skills are rewarded differently depending on the type of business.

For worker i in year t in a business of type $j \in \{B, A\}$, the model is

$$\ln y_{it} = \beta_{0j} + \beta_{1j} BX_{it} + \beta_{2j} BX_{it}^2 + \beta_{3j} SX_{it} + \beta_{4j} SX_{it}^2 + \beta_{5j} BSC_{it} + \beta_{6j} BSC_{it}^2 + W_{it}\beta_{Wj} + u_{it}, \tag{12}$$

where y_{it} is either annual business revenue or weekly earnings, and W_{it} contains the control variables described in Section 4.4.5: all baseline regressors except net family wealth and net family income, tenure in the current business and its square, and region and industry fixed effects (Panels B–E of Table 1). In the conceptual framework, the relationships between BX_{it} and SX_{it} and the outcome, which depend on both the linear and the quadratic terms, correspond to the skill-specific returns r_B^{BX}, r_B^{SX}, r_A^{BX}, and r_A^{SX}.

When the outcome in equation (12) is log annual revenue, I estimate the model using a Tobit, because revenue is top-coded in the NLSY79. When the outcome is log weekly earnings, I estimate the model by OLS. I estimate the earnings model on two samples. The first is the restricted sample of observations for which revenue is also available, so that revenue and earnings can be compared for the same businesses and years. The second is the full earnings sample, which shows whether the earnings results extend beyond the businesses with revenue information. Standard errors are clustered at the individual level in all models.

5.3 Selection on Observables

Workers with a strong taste for autonomy are likely to accumulate relatively large amounts of BX_{it} because they hope to own a business. If I cannot control for this taste, I cannot separate the direct (causal) effect of BX_{it} on entry into business ownership from the confounding effect of the taste for autonomy. This is a classic endogeneity problem. The same problem arises in the revenue and earnings models, where unobserved characteristics may be related to both the skills workers accumulate and how well their businesses perform.

My estimates rely on the assumption of selection on observables. Conditional on the baseline regressors, the remaining determinants of entry, business revenue, and earnings are assumed to be unrelated to the skill measures. Although this assumption cannot be

tested directly, I examine how the estimated marginal effects of the skill measures in the entry model change as I add groups of controls one at a time (Table 8). The most relevant groups for this check are those most closely related to preferences for business ownership: personality traits (including risk tolerance), whether a family member owned a business, and whether the respondent considers herself an entrepreneur. The last variable is added beyond the baseline regressors. If unobserved tastes are related to the skill measures in the same way as the observed taste proxies, small changes in the estimates when these proxies are added suggest that controlling for unobserved tastes would also have little effect on the estimates. Large changes would suggest that unobserved tastes may play a more important role (Altonji, Elder and Taber, 2005).

I also consider two alternatives to selection on observables for addressing the endogeneity problem, discussed in Appendix B. The first is individual fixed effects, which absorb time-invariant tastes. For entry, this approach is less informative because skills accumulate steadily with experience, leaving little within-worker variation after controlling for experience. Individual fixed effects are more useful for earnings, which are observed in both wage employment and business ownership. Comparing a worker's earnings across employment types helps identify how the earnings difference between business ownership and wage employment varies with her skills. I report these estimates in Appendix B.2. The second is a pair of instruments based on the industrial composition of the county where a worker starts her career. However, the instruments for BX_{it} and SX_{it} are highly correlated across counties (about 0.9), making it difficult to separate the effects of the two types of skills.

Although I hope to pursue alternative estimation methods in future work, the lack of strictly causal estimates of each effect of BX_{it} and SX_{it} is less concerning for two reasons. First, my comparison with conventional tests of the JOAT hypothesis focuses on which dimensions of work experience predict entry into business ownership, rather than on identifying the causal effects of each skill measure. Conventional tests also rely on associations conditional on observed characteristics. My question is what these associations reflect, such as whether the relationship between occupation count and entry reflects skill balance or job churning. Second, the most likely source of bias cannot explain why business skills predict entry in opposite directions for the two types of business ownership. Workers who plan to own a business may accumulate more business skills, which would bias the estimated effect of BX_{it} upward for entry into both types. This bias cannot explain the negative estimate for entry into basic business ownership. Instead, it would make the negative estimate smaller in magnitude.

5.4 Average Marginal Effects

To draw inferences from the estimated coefficients of each model, I compute average marginal effects (AMEs). For each observation, I compute the change in the predicted outcome from a marginal increase in a skill measure, holding all other variables at their observed values, and then average these changes over the sample. In the entry models, I express these effects as percentage changes in the predicted probability of each type of entry. In the revenue and earnings models, the outcomes are in logs, so the AMEs are changes in log points. AMEs facilitate the interpretation of the key relationships across specifications and models, given that I use a nonlinear model for entry and allow each skill measure to have a nonlinear relationship with each outcome. I standardize BX_{it}, SX_{it}, and BSC_{it} so that a one-unit change corresponds to a half-standard-deviation increase. This is approximately two years as a CEO for business skills and two and a half years for specialized skills in the occupation that uses a worker's specialization elements most intensively. I use half rather than a full standard deviation because a full one is nearly as large as the average level of business skills in my sample.

6 Findings

I first discuss the estimates of the competing-risks multinomial logit (MNL) model of entry into business ownership (Section 6.1) and then those of the revenue and earnings models for business owners (Section 6.2). The discussion focuses on the estimated marginal effects of business skills, specialized skills, and skill balance on each outcome, and on whether the estimates support my three innovations to the JOAT literature. Results of comparing my direct measure of skill balance with the occupation count is discussed in the entry model only. Because the estimated coefficients of these models are difficult to interpret, I focus on average marginal effects (AMEs), which are reported in Tables 7–9 and computed from the coefficient estimates in Appendix Tables A2, A3, and A5. As described in Section 5.4, each AME is the change in the predicted outcome from a half-standard-deviation increase in the measure.

6.1 Entry into Business Ownership

Before turning to the estimates, I restate the arguments they are designed to test. First, I argue that business skills and specialized skills, rather than balance across all skills, prepare workers for business ownership. Business skills should raise entry into advanced business ownership and lower entry into basic business ownership (Prediction 1), and specialized skills should lower entry into both types (Prediction 2). If the types of skills are what

matter rather than their balance, the balanced skill count should not predict entry into business ownership once business skills and specialized skills are controlled for. Second, because business skills are predicted to have opposite effects on entry into basic and advanced business ownership, pooling the two types can hide these relationships. Third, I replicate studies that pool business types and use the occupation count as a proxy for skill balance and examine whether its association with entry into business ownership reflects skill balance or job churning.

Table 7 reports AMEs from two models, both estimated on the same sample with the same baseline regressors. Panel A reports the MNL estimates for entry into basic and advanced business ownership, and Panel B reports the binary logit estimates for entry into either type. Each column is a separate specification. Columns 1 to 5 test whether the types of skills, rather than their balance, predict entry. Columns 1 to 3 include business skills and specialized skills, first separately and then together. Column 4 includes the balanced skill count on its own, and column 5 adds business skills and specialized skills to show whether skill balance still predicts entry once they are accounted for. Column 5 corresponds to equation (11) and is my main specification. Comparing the two panels shows whether the relationship between skills and entry depends on the type of business ownership, which pooling would hide. Finally, columns 6 to 8 compare my direct measure of skill balance with the occupation count, starting with column 6, which replicates the conventional test by including only the number of occupations held to date. Column 7 adds the three skill measures to show whether the number of occupations still predicts entry once skill balance is measured directly, and column 8 adds the number of involuntary job losses to show whether the occupation count partly reflects job churning.

Workers with more business skills are more likely to enter advanced business ownership. In column 5 of Panel A in Table 7, a half-standard-deviation increase in business skills, or about two years, is predicted raise the probability of entering advanced business ownership in a given year by 34.4 percent, compared with an average annual entry rate of 0.4 percent. In contrast, business skills are negatively associated with entry into basic business ownership. The same increase is predicted to lower the probability of entering basic business ownership by 14.4 percent, where the average annual entry rate is 1.4 percent. The estimates are similar across columns 1 and 3, which do not include the balanced skill count. This pattern supports my argument that business skills are rewarded more when running an advanced business than in wage employment, but less when running a basic one (Prediction 1). Pooling the two types hides this pattern. In the binary logit in Panel B, the estimated marginal effect of business skills on entry into either type of business is only 3.3 percent and not statistically significant, because the positive and negative associations cancel out.

Specialized skills are negatively associated with entry into business ownership, mainly basic business ownership. In column 5 of Panel A in Table 7, a half-standard-deviation increase in specialized skills, or about two and a half years, is predicted to lower the probability of entering basic business ownership by 13.1 percent. The same increase is also predicted to lower the probability of entering advanced business ownership, by 4.8 percent, but this estimate is smaller and not statistically significant. The binary logit in Panel B also predicts a 10.8 percent lower probability of entering either type. This does not mean that business owners do not need specialized skills. A dentist who opens her own practice, for example, still needs specialized knowledge of dentistry. However, these skills are likely to yield higher returns in wage employment than in business ownership, so workers with more specialized skills may face a greater opportunity cost when leaving wage employment (Prediction 2).

When the balanced skill count enters the entry model on its own, in column 4 of Panel A in Table 7, a half-standard-deviation increase is associated with a 7.6 percent higher probability of entering advanced business ownership, as the JOAT hypothesis predicts. Its estimate for basic business ownership is small, at 1.3 percent, and not statistically significant. Once business skills and specialized skills are taken into account as in column 5, skill balance no longer predicts entry into either type of business ownership. The AME of balanced skill count for advanced business ownership falls to 1.0 percent and is no longer statistically significant, and the AME for basic business ownership stays small, at 1.8 percent. Workers with more balanced skills tend to have more business skills, with a correlation of 0.21 in the entry sample (Table 6), so the association in column 4 likely reflects business skills rather than balance. Business skills, rather than skill balance, predict entry into advanced business ownership, and skill balance predicts entry into neither type.

My finding that skill balance does not predict entry into business ownership differs from the findings of many prior studies, most of which measure skill balance using the number of occupations or employers a worker has held (Section 2). Unlike the balanced skill count, which measures skill balance based on the skills used in each occupation, the occupation count is an indirect proxy that treats all occupations as equally valuable (Section 4.4). To reconcile these findings, I first replicate the conventional approach, pooling basic and advanced business ownership into a single outcome. Column 6 of Panel B in Table 7 shows that a half-standard-deviation increase in the number of occupations held to date predicts an 11.0 percent higher probability of entering business ownership, consistent with prior studies. However, the type-specific estimates in Panel A reveal that this results is driven almost entirely by basic business ownership. The same increase predicts a 14.8 percent higher probability of entering basic business ownership but only a 1.2 percent higher probability of entering advanced. Adding the three skill measures in column 7 leaves this pattern largely

unchanged. The estimate for basic business ownership falls to 12.9 percent but remains statistically significant, while the estimate for advanced business ownership stays small and statistically insignificant. Employer count, another common proxy, has a similar limitation: it increases whenever a worker changes employers, regardless of whether the new job uses different skills. Both counts may therefore capture job churning as well as skill balance, which I examine next.

Part of the occupation count's predictive power for entry into basic business ownership may reflect job churning rather than skill balance. As discussed inSection 2, the occupation count increases with any job mobility, including moves driven by poor performance, and job churning itself predicts entry into business ownership. To account for this, I add the number of involuntary job losses to the model in column 8 of Table 7. The two measures are correlated (0.48 in the entry sample (Table 6). Controlling for involuntary job losses reduces the estimated marginal effect of the number of occupations held to date on entry into basic business ownership by about one-third, from 12.9 percent in column 7 to 8.4 percent in column 8 of Panel A, although the estimate remains statistically significant. The number of involuntary job losses themselves predict a 11.3 percent higher probability of entering basic business ownership per half-standard-deviation increase, compared with 6.5 percent for advanced business ownership. These results suggest that job churning partly explains why the occupation count predicts entry into basic business ownership, even though the balanced skill count does not predict entry.

Because workers with a strong taste for business ownership may accumulate more business skills (Section 5.3), I examine how the estimated marginal effects change as I add groups of controls one at a time in Table 8, with the groups closest to tastes entering last. The estimated marginal effect of business skills on entry into advanced business ownership remains large and statistically significant in every column. It falls from 42.4 percent in column 1, which controls only for experience, to 34.4 percent in my main specification in column 7. The largest drop occurs when education and AFQT percentile scores are added, while personality traits and family business background, that might reflect tastes to some extent, lower the estimate only from 37.3 to 34.4 percent. Adding whether the respondent considers herself an entrepreneur lowers it further to 23.6 percent in column 8, but this question was asked only from 2010, when many respondents had already owned a business, so their answers may partly reflect that experience. For basic business ownership, the estimate becomes more negative as controls are added, reaching -14.4 percent in column 7. The estimate for the balanced skill count is more sensitive. For advanced business ownership, it falls from 12.3 percent in column 1 to 1.5 percent in column 3, once gender, race, and household characteristics are added, and stays small in the later columns. This pattern suggests that, among

workers with similar business and specialized skills, the association between skill balance and entry into advanced business ownership in columns 1 and 2 reflects differences in these characteristics rather than skill balance itself. These results cannot rule out confounding by unobserved tastes, but they are consistent with my assumption of selection on observables.

The entry results support each of my three innovations. First, the types of skills workers have accumulated, rather than the balance of their skills, predict entry. Business skills predict entry into advanced business ownership in particular, while the balanced skill count predicts entry into neither type once business and specialized skills are accounted for. Second, distinguishing basic from advanced business ownership matters. Pooling the two types hides the opposite associations of business skills with entry into each type, as well as the fact that the occupation count predicts entry only into basic business ownership. Third, the occupation count partly reflects job churning, and its association with entry into basic business ownership falls by about a third once involuntary job losses are accounted for. Together, these findings suggest that the support for the JOAT hypothesis in prior studies, which pool all business owners and measure skill balance with occupation counts, comes mainly from entry into basic business ownership and partly reflects job churning, rather than balanced skills.[23]

6.2 Business Revenue and Earnings

I next examine whether the same skills predict business revenue and the owner's weekly earnings after entry. While the entry decisions depends on how a skill is rewarded in business ownership relative to wage employment, business revenue and earnings depend on how it is rewarded within business ownership. So, a skill that reduces the probability of entry may still predict higher payoffs among those who enter. For this reason, both business and specialized skills may predict higher revenue and earnings in either type of business ownership if they are valuable in running a business. If business skills are particularly valuable in advanced business ownership, we would expect their returns to be greater in advanced than in basic business ownership (Prediction 3).

Table 9 reports the estimated marginal effects separately for basic business ownership in Panel A and advanced business ownership in Panel B. Column 1 reports the Tobit estimates for log annual revenue, and columns 2 and 3 report the OLS estimates for log weekly earnings.

[23]My findings also help reconcile evidence in the JOAT literature that has been difficult to explain. Lechmann and Schnabel (2014) find that occupation counts predict entry into solo self-employment but not into self-employment with employees, while Åstebro and Thompson (2011) find that workers with more occupations tend to earn less in self-employment. Neither finding is consistent with the JOAT hypothesis, but both are consistent with my results if solo self-employment more closely resembles basic business ownership, where occupation counts partly reflect job churning and earnings are substantially lower than in advanced business ownership (Table 5).

Column 2 uses the revenue sample, which covers the same businesses and years as column 1, while column 3 uses the full earnings sample. Because the outcomes are in logs, the estimates are in log points, and in the text I convert each estimate b to a percent change, $100(e^b - 1)$.

Business skills predict higher business revenue in both types of business ownership, and the estimate is larger in advanced business ownership. In column 1 of Panel B, a half-standard-deviation increase in business skills is associated with 52.0 percent higher revenue. The estimates for earnings are smaller, at 9.9 percent in column 2 and 14.5 percent in column 3. In basic business ownership, the same increase predicts 28.7 percent higher revenue in column 1 of Panel A. The estimates for earnings are also positive, at 14.8 percent in column 2 and 4.2 percent in column 3, although neither is statistically significant. In column 3, which uses the full earnings sample, the earnings estimate is larger in advanced than in basic business ownership. In both types, the estimate is larger for revenue than for earnings. One possible explanation is that owners with more business skills grow their businesses by reinvesting profits rather than taking more income home. If so, earnings understate how much business skills contribute to business performance.

Specialized skills predict higher business revenue and earnings in basic business ownership, even though they lower the predicted probability of entering it. In Panel A, a half-standard-deviation increase in specialized skills is predicted to raise revenue by 17.6 percent in column 1 and earnings by 15.0 and 20.1 percent in columns 2 and 3. In Panel B, the estimates for advanced business ownership are smaller and less precise, at 6.9 percent for revenue and 6.3 percent for earnings in both columns 2 and 3, and none is statistically significant. Unlike business skills, specialized skills have similar estimates for revenue and earnings, which suggests that their contribution to business performance translates more directly into owners' earnings.

The balanced skill count shows little relationship with business revenue or earnings. In Panel B, none of its estimates is statistically significant. In Panel A, a half-standard-deviation increase predicts 7.0 percent higher revenue in column 1 and 3.4 percent higher earnings in column 3, significant only at the 10 and 5 percent levels, and both estimates are less than half the size of the corresponding estimates for specialized skills. Together with the entry results, these estimates give little support to the JOAT hypothesis that balance across all skills prepares workers for business ownership.

In sum, the revenue and earnings results support my first two innovations. Business skills and specialized skills, rather than skill balance, predict business revenue and earnings, and how strongly each predicts them depends on the type of business ownership (Prediction 3). Business skills are more strongly associated with revenue and earnings in advanced business ownership, and specialized skills in basic business ownership.

7 Conclusion

I study what kind of work experience prepares someone to start a business. I extend the JOAT literature in three ways. First, I argue that what prepares a worker for business ownership is not balance across all skills, but business skills and specialized skills relevant to the business. Second, I distinguish between basic and advanced business ownership, where advanced businesses are incorporated and/or have higher start-up capital. Third, I compare the occupation count, the conventional proxy for skill balance, with a direct measure of skill balance. Using nearly 40 years of NLSY79 work histories linked to O*NET, I estimate a competing-risks model of entry into each type of business ownership, as well as models of business revenue and earnings after entry.

My findings support each of these innovations. First, business skills predict entry into advanced business ownership, while my direct measure of skill balance adds no predictive power for entry into either type of business. This suggests that owners of advanced businesses are jacks of *business* trades rather than jacks of all trades. Second, separating basic from advanced business ownership reveals differences that pooled analyses hide. Business skills predict entry into advanced business ownership but deter entry into basic business ownership, and they are more strongly associated with business revenue and earnings in advanced businesses, while specialized skills are more strongly associated with revenue and earnings in basic businesses. Third, part of what the occupation count measures is job churning. It predicts entry only into basic business ownership, and about a third of this association disappears when I control for involuntary job losses. Together, these findings suggest that conventional tests of the JOAT hypothesis, which combine all types of business owners and use occupation counts as a proxy for skill balance, may attribute patterns of business ownership entry to skill balance that actually reflect business skills and job churning.

Consistent with my conceptual framework, the skills that deter entry into business ownership may still predict higher business revenue and earnings after entry. This is because entry depends on how the returns to a skill in business ownership compare with its returns in wage employment, whereas revenue and earnings depend on its returns within business ownership. Specialized skills show this difference. Workers with more specialized skills may give up more when they leave wage employment and are less likely to enter basic business ownership in particular, but these skills are predictive of higher revenue and earnings among basic business owners.

Knowing which skills prepare workers for business ownership can help target support to their needs, and the right support depends on the type of business. For advanced business ownership, experience in occupations that involve managing people, budgets, and customers

builds the business skills that predict entry, whereas experience across many unrelated jobs does not. SBDCs help workers build these skills before they start a business, and programs such as the SBA's THRIVE Emerging Leaders and Empower to Grow offer management training to existing business owners. Owners of basic businesses need a different kind of support. Owners of basic businesses need a different kind of support. Because specialized skills are more strongly associated with business revenue and earnings among basic business owners, support for these owners may be more effective if it builds on their occupation-specific expertise, for example through mentoring from people in the same line of work, as SCORE offers. Since involuntary job losses predict entry into basic business ownership, programs that serve displaced workers may be a natural place to offer this support. Whether training can build the same skills that workers accumulate on the job remains an open question, which I return to below.

This analysis points to several directions for further work. The first is about interpretation. My estimates rely on selection on observables. Although they are largely robust to adding the rich set of controls available in the NLSY79, tastes I do not observe may still shape both the skills a worker accumulates and her decision to start a business. An instrument that shifts the skills a worker accumulates but affects business ownership only through those skills would address this problem. The second is about whose skills matter. Owners can obtain skills they lack by forming a team, which I do not observe in the NLSY79, and data on founding teams would show whether the skills a business needs must be held by the owner or can be supplied by other members of the team. The third concerns measurement. My skill measures rely on occupation-level averages, which may mask differences in the tasks workers perform within the same occupation, and individual-level task data would capture this variation. The fourth is whether these skills can be taught. Testing whether business skills and specialized skills developed through training programs carry the same returns as those accumulated on the job would show whether training can substitute for work experience in preparing workers for business ownership.

The JOAT hypothesis asks how balanced an owner's skills are. My findings suggest asking instead which skills. What prepares a worker for business ownership, and for success in it, is not skill balance, but the right type of skills for the type of business she runs.

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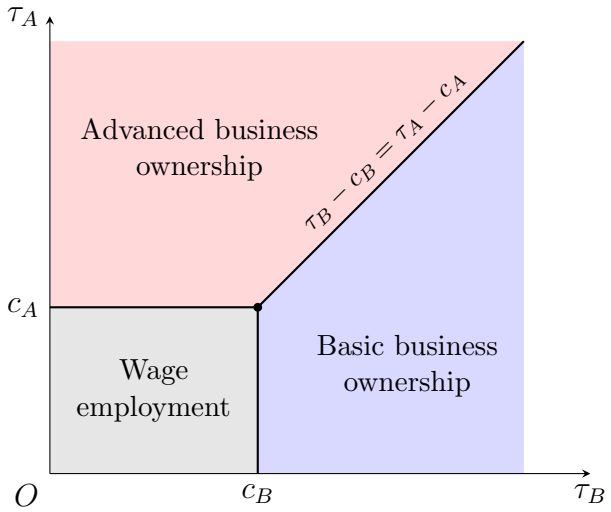
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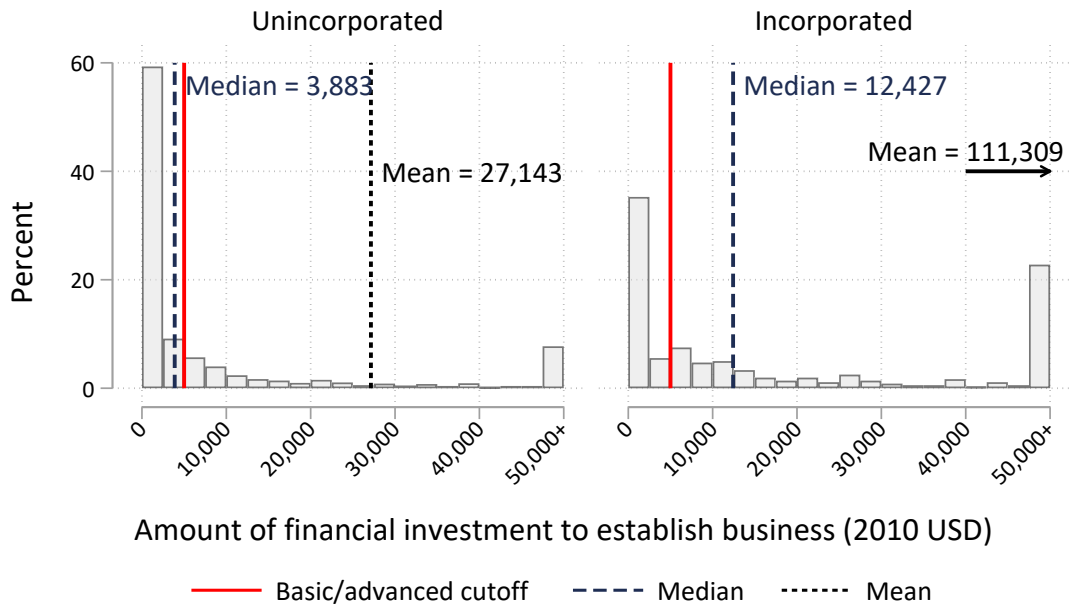
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Figure 1: The Worker's Choice Between Employment Options as a Function of Tastes for Business Ownership



Note: As defined in Section 3, τ_B and τ_A are the worker's tastes for basic and advanced business ownership, respectively, while c_B and c_A are the opportunity cost of each business option. Each shaded region identifies the taste (τ_B, τ_A) combinations that lead to the option indicated, given the opportunity costs c_B and c_A.

Figure 2: Distribution of Start-Up Capital by Incorporation Status



Note: This figure shows the distribution of start-up capital at the start of each business, separately for incorporated and unincorporated businesses. The red line indicates the $5,000 cutoff (equal to the median across all businesses with positive start-up capital) that I use to distinguish basic from advanced businesses.

Table 1: Means and Standard Deviations for All Regressors in the Five Estimation Samples

	Business Ownership	Business Revenue		Weekly Earnings	
	Entry	Basic	Advanced	Basic	Advanced
	(1)	(2)	(3)	(4)	(5)
Panel A: Key Independent Variables					
O*NET Business skills (years)	4.81	7.56	8.69	4.32	6.73
	(4.46)	(5.47)	(5.96)	(4.40)	(5.46)
O*NET Specialized skills (years)	4.92	6.17	7.06	3.74	5.49
	(5.34)	(5.52)	(5.74)	(4.30)	(5.13)
O*NET Balanced skill count	28.55	28.77	28.96	28.60	28.88
	(0.81)	(0.65)	(0.57)	(0.77)	(0.65)
Number of occupations held to date	4.58	–	–	–	–
	(2.44)				
Number of involuntary job losses to date	2.01	–	–	–	–
	(2.53)				
Panel B: Baseline Regressors—Work Experience					
Potential experience (years)	19.18	24.18	20.82	15.40	16.76
	(11.71)	(9.23)	(8.50)	(9.58)	(8.96)
Actual experience (years)	14.23	18.52	18.07	11.15	14.46
	(10.98)	(9.41)	(8.73)	(8.71)	(8.76)
Number of past business ownership spells	0.29	2.53	2.13	1.56	1.76
	(0.69)	(3.00)	(2.77)	(2.24)	(2.41)
Tenure in the business (years)	–	5.40	10.22	3.05	7.40
		(7.87)	(10.54)	(5.50)	(8.98)
Panel C: Baseline Regressors—Individual Characteristics					
1 if female[a]	0.53	0.47	0.25	0.46	0.27
Race/ethnicity[a]					
1 if Hispanic	0.19	0.16	0.16	0.16	0.16
1 if Black	0.32	0.26	0.18	0.24	0.16
1 if neither Black nor Hispanic	0.49	0.58	0.66	0.61	0.68
Marital status					
1 if never married	0.29	0.16	0.09	0.22	0.12
1 if currently married	0.56	0.53	0.70	0.56	0.70
1 if previously married	0.15	0.31	0.21	0.22	0.18
1 if cohabiting with an unmarried partner	0.10	0.04	0.04	0.06	0.04
1 if youngest child is aged six or younger	0.24	0.05	0.07	0.23	0.18
Education					
1 if less than high school	0.13	0.07	0.05	0.12	0.06
1 if high school graduate	0.45	0.39	0.36	0.43	0.39
1 if some college	0.24	0.22	0.23	0.22	0.24
1 if college or above	0.19	0.33	0.36	0.23	0.31
AFQT percentile score[a]	40.92	51.18	52.87	46.51	52.10

Continued on next page

Table 1 (continued)

	Business Ownership	Business Revenue		Weekly Earnings	
	Entry	Basic	Advanced	Basic	Advanced
	(1)	(2)	(3)	(4)	(5)
	(28.53)	(29.87)	(28.04)	(29.44)	(27.63)
Risk tolerance					
1 if least risk tolerant	0.52	0.45	0.41	0.46	0.46
1 if less risk tolerant	0.14	0.20	0.22	0.15	0.16
1 if more risk tolerant	0.14	0.12	0.11	0.14	0.13
1 if most risk tolerant	0.20	0.22	0.26	0.24	0.25
1 if unknown	0.00	0.01	0.00	0.01	0.00
Sociability at age six[a]					
1 if extremely shy	0.18	0.15	0.12	0.14	0.12
1 if somewhat shy	0.47	0.49	0.43	0.52	0.45
1 if somewhat outgoing	0.24	0.24	0.31	0.23	0.28
1 if extremely outgoing	0.11	0.12	0.15	0.12	0.15
Rotter locus of control score (4–16)	7.85	7.13	6.86	7.34	7.01
	(2.37)	(2.12)	(2.16)	(2.13)	(2.14)
Openness to Experience (TIPI, 1-7)[a]	5.23	5.32	5.39	5.31	5.35
	(1.57)	(1.44)	(1.44)	(1.34)	(1.41)
Extraversion (TIPI, 1–7)[a]	3.32	2.97	3.04	3.02	3.03
	(1.85)	(1.67)	(1.73)	(1.63)	(1.71)
1 if considers oneself an entrepreneur[a,b]	0.15	–	–	–	–
Panel D: Baseline Regressors—Family Background					
Net family wealth (thousands of $)	158.29	–	–	–	–
	(394.36)				
Net family income (thousands of $)	64.20	–	–	–	–
	(79.59)				
1 if father had at least some college[a]	0.19	0.29	0.30	0.25	0.30
1 if father was a professional or manager in 1979[a]	0.16	0.25	0.30	0.22	0.30
1 if any family member owned a business[a]	0.26	0.35	0.43	0.34	0.44
Panel E: Region and Industry					
Region					
1 if Northeast	–	0.14	0.14	0.15	0.14
1 if North Central	–	0.25	0.26	0.25	0.27
1 if South	–	0.39	0.37	0.37	0.36
1 if West	–	0.20	0.21	0.22	0.21
1 if unknown	–	0.01	0.02	0.01	0.02
Industry					
1 if mining and construction	–	0.16	0.17	0.16	0.17
1 if manufacturing	–	0.03	0.07	0.03	0.08

Continued on next page

Table 1 (continued)

	Business Ownership	Business Revenue		Weekly Earnings	
	Entry	Basic	Advanced	Basic	Advanced
	(1)	(2)	(3)	(4)	(5)
1 if transportation, communications, and public utilities	–	0.04	0.08	0.03	0.08
1 if wholesale and retail trade	–	0.08	0.15	0.09	0.17
1 if services	–	0.32	0.26	0.40	0.27
1 if professional services	–	0.28	0.17	0.19	0.15
1 if other	–	0.09	0.08	0.09	0.08
Number of person-year observations	251,214	2,430	2,465	10,383	5,938
Number of individuals	7,116	938	832	2,370	1,103

Note: Summary statistics are computed over the person-years of each estimation sample. In the business revenue and weekly earnings samples, the skill measures and experience are measured at the start of each business. See Section 4 for details. A dash means that the regressor is not included in the model for that outcome.

[a] Time-invariant; all other variables are time-varying.

[b] I use this variable only in column 8 of Table 8.

Table 2: Distribution of Individuals by Number and Type of Business Ownership Entries

	N. individuals	Share (%)
Panel A: Number of Entries into Business Ownership		
0	4,166	58.5
1	1,838	25.8
2	761	10.7
3	240	3.4
4	81	1.1
5	20	0.3
6	7	0.1
7	1	0.0
8	2	0.0
Total	7,116	100.0
Panel B: Types of Business Ownership Entered		
Never entered	4,166	58.5
Basic only	1,989	28.0
Advanced only	596	8.4
Both	365	5.1
Total	7,116	100.0

Table 3: The Eight Most Common Occupations by Type of Business

	Basic Business	Advanced Business
1	Childcare Workers	Managers, All Other
2	Landscaping and Groundskeeping Workers	Sales Workers
3	Sales Workers	Driver/Sales Workers and Truck Drivers
4	Managers, All Other	First-Line Supervisors of Retail Sales Workers
5	Maids and Housekeeping Cleaners	Carpenters
6	Carpenters	Bookkeeping, Accounting, and Auditing Clerks
7	Painters, Construction and Maintenance	Chief Executives
8	Construction Laborers	Food Service Managers

Note: The eight most common occupations are listed in order of frequency, based on 3,443 basic businesses and 1,108 advanced businesses owned by individuals in the entry sample. Each business is assigned the owner's 6-digit SOC occupation at entry.

Table 4: Means and Standard Deviations for Selected Characteristics

	No Entry (1)	Basic (2)	Advanced (3)
Age (years)	38.93	34.99	38.50
	(11.72)	(10.77)	(9.84)
Potential experience (years)	19.24	15.46	18.09
	(11.72)	(10.68)	(10.07)
Actual experience (years)	14.27	10.86	15.33
	(11.00)	(9.61)	(9.81)
O*NET Business skills (years)	4.82	3.60	5.96
	(4.47)	(3.87)	(4.69)
O*NET Business skills per year of actual experience[a]	0.32	0.31	0.38
	(0.11)	(0.11)	(0.13)
O*NET Specialized skills (years)	4.94	3.24	5.10
	(5.36)	(3.92)	(4.52)
O*NET Specialized skills per year of actual experience[a]	0.32	0.30	0.33
	(0.20)	(0.18)	(0.19)
O*NET Balanced skill count	28.55	28.56	28.83
	(0.81)	(0.78)	(0.69)
Number of occupations held to date	4.58	4.77	4.96
	(2.44)	(2.56)	(2.44)
Number of occupations held per year of actual experience[a]	0.53	0.71	0.48
	(0.54)	(0.60)	(0.47)
Number of involuntary job losses per year of actual experience[a]	0.22	0.32	0.17
	(0.37)	(0.45)	(0.26)
Number of past business ownership spells per year of actual experience[a]	0.03	0.07	0.05
	(0.08)	(0.14)	(0.09)
Education			
1 if less than high school	0.13	0.16	0.08
1 if high school graduate	0.45	0.44	0.37
1 if some college	0.24	0.22	0.24
1 if college or above	0.19	0.18	0.31
AFQT percentile score	40.84	43.05	50.99
	(28.52)	(28.93)	(28.22)
Number of person-year observations	246,645	3,463	1,106

Note: Summary statistics are calculated over the person-years of the entry sample. Columns 2 and 3 use the last person-year before entry into basic and advanced business ownership, respectively. Column 1 uses all other person-years, in which no entry occurs in the following year.

[a] Per-year measures exclude person-years with less than one year of actual experience.

Table 5: Means and Standard Deviations for Business Characteristics by Business Type

	Basic (1)	Advanced (2)
Annual revenue, excluding top-coded values (thousands of $)[a]	63.32	215.48
	(291.10)	(439.64)
1 if revenue is top-coded[a]	0.01	0.09
Weekly earnings ($)	443.47	843.63
	(717.63)	(1,198.74)
Hours worked per week	31.91	40.86
	(19.82)	(20.95)
Business duration (years)	3.93	7.61
	(5.50)	(7.74)
1 if business lasted less than 1 year	0.30	0.13
1 if business lasted 5 years or more	0.23	0.49
1 if business is ongoing at the last interview	0.12	0.32
Occupation group		
1 if management, business, science, and arts	0.25	0.42
1 if service	0.34	0.11
1 if sales and office	0.14	0.21
1 if natural resources, construction, and maintenance	0.19	0.15
1 if production, transportation, and material moving	0.08	0.11
Number of businesses, revenue sample	1,221	1,027
Number of businesses, earnings sample	4,118	1,554

[a] These statistics are from the revenue sample; all other rows are from the earnings sample.

Table 6: Partial Correlations of the Skill and Count Measures

	(1)	(2)	(3)	(4)	(5)
Panel A: Entry Sample, Person-Years					
(1) O*NET Business skills (years)	1.000				
(2) O*NET Specialized skills (years)	0.396***	1.000			
(3) O*NET Balanced skill count	0.214***	-0.086***	1.000		
(4) Number of occupations held to date	-0.105***	-0.222***	0.249***	1.000	
(5) Number of involuntary job losses to date	-0.241***	-0.167***	0.091***	0.476***	1.000
Number of person-year observations = 251,214					
Panel B: Basic Businesses in the Earnings Sample, at Business Start					
(1) O*NET Business skills (years)	1.000				
(2) O*NET Specialized skills (years)	0.451***	1.000			
(3) O*NET Balanced skill count	0.145***	-0.026*	1.000		
(4) Number of occupations held to date	-0.134***	-0.194***	0.252***	1.000	
(5) Number of involuntary job losses to date	-0.129***	-0.125***	0.128***	0.358***	1.000
Number of businesses = 4,118					
Panel C: Advanced Businesses in the Earnings Sample, at Business Start					
(1) O*NET Business skills (years)	1.000				
(2) O*NET Specialized skills (years)	0.559***	1.000			
(3) O*NET Balanced skill count	0.158***	0.012	1.000		
(4) Number of occupations held to date	-0.193***	-0.262***	0.182***	1.000	
(5) Number of involuntary job losses to date	-0.232***	-0.208***	0.067***	0.392***	1.000
Number of businesses = 1,554					

Note: Correlations control for actual experience within each panel. *p < .10; **p < .05; ***p < .01.

Table 7: Estimated Average Marginal Effects of Key Regressors on the Probability of Business Ownership Entry (Based on MNL and Logit Estimates Shown in Table A2)

	Dependent Variable: Entry into Business Ownership							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Multinomial Logit								
Basic Business Ownership (Unconditional Prob. = 0.014)								
Std. O*NET Business skills	-0.192***		-0.135***		-0.144***		-0.122***	-0.069
	(0.040)		(0.041)		(0.043)		(0.044)	(0.045)
Std. O*NET Specialized skills		-0.152***	-0.136***		-0.131***		-0.108***	-0.114***
		(0.021)	(0.022)		(0.022)		(0.023)	(0.023)
Std. O*NET Balanced skill count				0.013	0.018		-0.010	-0.012
				(0.014)	(0.015)		(0.015)	(0.015)
Std. Number of occupations held to date						0.148***	0.129***	0.084***
						(0.013)	(0.015)	(0.017)
Std. Number of involuntary job losses to date								0.113***
								(0.020)
Advanced Business Ownership (Unconditional Prob. = 0.004)								
Std. O*NET Business skills	0.345***		0.350***		0.344***		0.340***	0.362***
	(0.063)		(0.066)		(0.068)		(0.069)	(0.070)
Std. O*NET Specialized skills		0.010	-0.049		-0.048		-0.050	-0.052
		(0.032)	(0.032)		(0.032)		(0.032)	(0.032)
Std. O*NET Balanced skill count				0.076***	0.010		0.014	0.013
				(0.024)	(0.026)		(0.026)	(0.026)
Std. Number of occupations held to date						0.012	-0.015	-0.037
						(0.023)	(0.026)	(0.028)
Std. Number of involuntary job losses to date								0.065**
								(0.030)

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Table 7 (continued)

	Dependent Variable: Entry into Business Ownership							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel B: Binary Logit								
Any Business Ownership (Unconditional Prob. = 0.018)								
Std. O*NET Business skills	-0.006		0.036		0.033		0.052	0.098***
	(0.034)		(0.035)		(0.036)		(0.037)	(0.038)
Std. O*NET Specialized skills		-0.101***	-0.112***		-0.108***		-0.093***	-0.098***
		(0.018)	(0.019)		(0.019)		(0.019)	(0.019)
Std. O*NET Balanced skill count				0.028**	0.011		-0.009	-0.011
				(0.012)	(0.013)		(0.013)	(0.013)
Std. Number of occupations held to date						0.110***	0.091***	0.052***
						(0.012)	(0.013)	(0.015)
Std. Number of involuntary job losses to date								0.101***
								(0.017)
Number of individuals	7,116	7,116	7,116	7,116	7,116	7,116	7,116	7,116
Number of person-year observations	251,214	251,214	251,214	251,214	251,214	251,214	251,214	251,214

Note: In both panels, the reference category is staying out of business ownership. Estimated marginal effects are the average percentage change in the predicted probability of entry from a half-standard-deviation increase in each skill and count measure. See Table 1 for the full list of covariates included. Standard errors clustered at the individual level are in parentheses. *p < .10; **p < .05; ***p < .01.

Table 8: Sensitivity to Additional Controls: Estimated Average Marginal Effects of Key Regressors on the Probability of Business Ownership Entry (Based on MNL Estimates Shown in Table A3)

	Dependent Variable: Entry into Business Ownership							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Multinomial Logit								
Basic Business Ownership (Unconditional Prob. = 0.014)								
Std. O*NET Business skills	-0.066	-0.085**	-0.087**	-0.110***	-0.122***	-0.139***	-0.144***	-0.177***
	(0.041)	(0.040)	(0.040)	(0.043)	(0.043)	(0.043)	(0.043)	(0.043)
Std. O*NET Specialized skills	-0.127***	-0.138***	-0.132***	-0.134***	-0.133***	-0.132***	-0.131***	-0.118***
	(0.022)	(0.022)	(0.022)	(0.022)	(0.022)	(0.022)	(0.022)	(0.022)
Std. O*NET Balanced skill count	0.052***	0.041***	0.023	0.020	0.021	0.020	0.018	0.008
	(0.015)	(0.014)	(0.015)	(0.015)	(0.015)	(0.015)	(0.015)	(0.014)
Advanced Business Ownership (Unconditional Prob. = 0.004)								
Std. O*NET Business skills	0.424***	0.426***	0.492***	0.402***	0.373***	0.351***	0.344***	0.236***
	(0.064)	(0.065)	(0.065)	(0.068)	(0.068)	(0.068)	(0.068)	(0.066)
Std. O*NET Specialized skills	-0.074**	-0.086***	-0.043	-0.047	-0.045	-0.050	-0.048	-0.028
	(0.031)	(0.033)	(0.033)	(0.032)	(0.032)	(0.032)	(0.032)	(0.034)
Std. O*NET Balanced skill count	0.123***	0.108***	0.015	0.013	0.013	0.014	0.010	0.003
	(0.026)	(0.026)	(0.027)	(0.026)	(0.026)	(0.026)	(0.026)	(0.025)
Potential and actual experience	✓	✓	✓	✓	✓	✓	✓	✓
Past business ownership spells		✓	✓	✓	✓	✓	✓	✓
Gender, race, and household composition			✓	✓	✓	✓	✓	✓
Education and AFQT				✓	✓	✓	✓	✓
Wealth, income, and father's background					✓	✓	✓	✓
Risk tolerance and personality						✓	✓	✓
Family member owned a business							✓	✓
Considers oneself an entrepreneur								✓
Number of individuals	7,116	7,116	7,116	7,116	7,116	7,116	7,116	7,116
Number of person-year observations	251,214	251,214	251,214	251,214	251,214	251,214	251,214	251,214

Note: The reference category is staying out of business ownership. Estimated marginal effects are the average percentage change in the predicted probability of entry from a half-standard-deviation increase in each skill measure. Columns 2–8 progressively add the control groups indicated at the bottom of the table. See Table 1 for the full list of covariates included. Standard errors clustered at the individual level are in parentheses. *p < .10; **p < .05; ***p < .01.

Table 9: Estimated Average Marginal Effects of Business Skills, Specialized Skills, and Skill Balance on Business Revenue and Weekly Earnings in Business Ownership (Based on Estimates Shown in Table A5)

	Revenue Sample		Earnings Sample
	Log Annual Revenue (Tobit)	Log Weekly Earnings (OLS)	Log Weekly Earnings (OLS)
	(1)	(2)	(3)
Panel A: Basic Business Ownership			
Std. O*NET Business skills	0.252**	0.138	0.041
	(0.108)	(0.106)	(0.081)
Std. O*NET Specialized skills	0.162***	0.140***	0.183***
	(0.048)	(0.054)	(0.043)
Std. O*NET Balanced skill count	0.068*	0.041	0.033**
	(0.042)	(0.043)	(0.016)
Tenure in the business	0.153***	0.131***	0.097***
	(0.014)	(0.014)	(0.008)
Number of person-year observations	2,430	2,430	10,383
Number of businesses	1,221	1,221	4,118
Mean business revenue or earnings	56,859	412	514
Panel B: Advanced Business Ownership			
Std. O*NET Business skills	0.419***	0.094	0.135*
	(0.115)	(0.093)	(0.078)
Std. O*NET Specialized skills	0.067	0.061	0.061
	(0.068)	(0.061)	(0.049)
Std. O*NET Balanced skill count	0.013	0.044	0.036
	(0.062)	(0.046)	(0.029)
Tenure in the business	0.106***	0.100***	0.089***
	(0.011)	(0.010)	(0.007)
Number of person-year observations	2,465	2,465	5,938
Number of businesses	1,027	1,027	1,554
Mean business revenue or earnings	215,578	1,088	1,088

Note: Estimated marginal effects correspond to a half-standard-deviation increase in each skill and are in log points. Mean revenue in column 1 excludes top-coded reports. See Table 1 for the full list of covariates included. Standard errors clustered at the individual level are in parentheses. *p < .10; **p < .05; ***p < .01.

Appendix A: Additional Tables

Table A1: Classification of O*NET Skill and Knowledge Elements by Skill Type

Skill Type	Skill Elements	Knowledge Elements
Core Business	Service Orientation	Economics and Accounting
		Sales and Marketing
		Customer and Personal Service
		Production and Processing
Managerial	Management of Financial Resources	Administration and Management
	Management of Material Resources	Personnel and Human Resources
	Management of Personnel Resources	
Interpersonal	Social Perceptiveness	
	Coordination	
	Persuasion	
	Negotiation	
	Instructing	
Other	Reading Comprehension	Clerical
	Active Listening	Transportation
	Writing	Food Production
	Speaking	Computers and Electronics
	Mathematics	Engineering and Technology
	Science	Design
	Critical Thinking	Building and Construction
	Active Learning	Mechanical
	Learning Strategies	Mathematics
	Monitoring	Physics
	Complex Problem Solving	Chemistry
	Operations Analysis	Biology
	Technology Design	Psychology
	Equipment Selection	Sociology and Anthropology
	Installation	Geography
	Programming	Medicine and Dentistry
	Operation Monitoring	Therapy and Counseling
	Operation and Control	Education and Training
	Equipment Maintenance	English Language
	Troubleshooting	Foreign Language
	Repairing	Fine Arts
	Quality Control Analysis	History and Archeology
	Judgment and Decision Making	Philosophy and Theology
	Systems Analysis	Public Safety and Security
	Systems Evaluation	Law and Government
	Time Management	Telecommunications
		Communications and Media

Note: This table shows my classification of all 68 O*NET skill and knowledge elements as core business, managerial, interpersonal, or other. I use the 15 elements not classified as "other" to construct my business skills measure.

Table A2: MNL and Logit Coefficient Estimates for Entry into Business Ownership

	Dependent Variable: Entry into Business Ownership							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Multinomial Logit								
Basic Business Ownership (Unconditional Prob. = 0.014)								
Std. O*NET Business skills	-0.195***		-0.138***		-0.146***		-0.124***	-0.069
	(0.041)		(0.042)		(0.044)		(0.045)	(0.046)
Std. O*NET Business skills^2	0.022***		0.028***		0.028***		0.025***	0.021***
	(0.005)		(0.006)		(0.006)		(0.006)	(0.006)
Std. O*NET Specialized skills		-0.154***	-0.138***		-0.132***		-0.109***	-0.115***
		(0.022)	(0.022)		(0.023)		(0.024)	(0.024)
Std. O*NET Specialized skills^2		-0.004	-0.012*		-0.012		-0.012	-0.012
		(0.006)	(0.008)		(0.007)		(0.007)	(0.007)
Std. O*NET Balanced skill count				0.014	0.018		-0.010	-0.012
				(0.014)	(0.015)		(0.015)	(0.015)
Std. O*NET Balanced skill count^2				-0.005*	-0.004		-0.004	-0.004
				(0.003)	(0.003)		(0.003)	(0.003)
Std. Number of occupations held to date						0.151***	0.132***	0.086***
						(0.014)	(0.016)	(0.018)
Std. Number of occupations held to date^2						-0.011***	-0.010**	-0.009**
						(0.004)	(0.004)	(0.004)
Std. Number of involuntary job losses to date								0.115***
								(0.020)
Std. Number of involuntary job losses to date^2								-0.008**
								(0.004)
Advanced Business Ownership (Unconditional Prob. = 0.004)								
Std. O*NET Business skills	0.345***		0.350***		0.344***		0.340***	0.363***
	(0.064)		(0.066)		(0.069)		(0.069)	(0.071)
Std. O*NET Business skills^2	-0.031***		-0.013		-0.013		-0.013	-0.015
	(0.010)		(0.011)		(0.011)		(0.011)	(0.011)
Std. O*NET Specialized skills		0.009	-0.049		-0.048		-0.050	-0.052
		(0.032)	(0.033)		(0.033)		(0.033)	(0.033)
Std. O*NET Specialized skills^2		-0.042***	-0.048***		-0.047***		-0.048***	-0.048***
		(0.010)	(0.011)		(0.011)		(0.011)	(0.012)
Std. O*NET Balanced skill count				0.077***	0.010		0.014	0.013
				(0.024)	(0.026)		(0.026)	(0.026)
Std. O*NET Balanced skill count^2				0.007	-0.002		-0.001	-0.002
				(0.005)	(0.007)		(0.007)	(0.007)
Std. Number of occupations held to date						0.014	-0.013	-0.036
						(0.023)	(0.026)	(0.028)
Std. Number of occupations held to date^2						-0.003	-0.001	-0.000
						(0.006)	(0.007)	(0.007)
Std. Number of involuntary job losses to date								0.067**
								(0.030)
Std. Number of involuntary job losses to date^2								-0.006
								(0.004)

Continued on next page

Table A2 (continued)

	Dependent Variable: Entry into Business Ownership							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel B: Binary Logit								
Any Business Ownership (Unconditional Prob. = 0.018)								
Std. O*NET Business skills	-0.007		0.036		0.033		0.053	0.099**
	(0.034)		(0.035)		(0.037)		(0.038)	(0.039)
Std. O*NET Business skills^2	0.005		0.015***		0.015***		0.013**	0.009
	(0.005)		(0.005)		(0.005)		(0.006)	(0.006)
Std. O*NET Specialized skills		-0.102***	-0.113***		-0.109***		-0.094***	-0.099***
		(0.019)	(0.019)		(0.019)		(0.020)	(0.020)
Std. O*NET Specialized skills^2		-0.014***	-0.023***		-0.022***		-0.022***	-0.022***
		(0.005)	(0.006)		(0.006)		(0.006)	(0.006)
Std. O*NET Balanced skill count				0.028**	0.011		-0.009	-0.011
				(0.012)	(0.013)		(0.013)	(0.013)
Std. O*NET Balanced skill count^2				-0.002	-0.004		-0.004	-0.004
				(0.003)	(0.003)		(0.003)	(0.003)
Std. Number of occupations held to date						0.112***	0.093***	0.053***
						(0.012)	(0.014)	(0.015)
Std. Number of occupations held to date^2						-0.009***	-0.008**	-0.008**
						(0.003)	(0.004)	(0.004)
Std. Number of involuntary job losses to date								0.103***
								(0.018)
Std. Number of involuntary job losses to date^2								-0.008**
								(0.003)
Number of individuals	7,116	7,116	7,116	7,116	7,116	7,116	7,116	7,116
Number of person-year observations	251,214	251,214	251,214	251,214	251,214	251,214	251,214	251,214

Note: The models are those in Table 7, with the same controls. All reported measures are standardized so that a one-unit increase equals half a standard deviation. Standard errors clustered at the individual level are in parentheses. *p < .10; **p < .05; ***p < .01.

Table A3: Sensitivity to Additional Controls: MNL Coefficient Estimates for Entry into Business Ownership

	Dependent Variable: Entry into Business Ownership							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Multinomial Logit								
Basic Business Ownership (Unconditional Prob. = 0.014)								
Std. O*NET Business skills	-0.067	-0.086**	-0.087**	-0.111**	-0.124***	-0.141***	-0.146***	-0.181***
	(0.042)	(0.040)	(0.041)	(0.043)	(0.044)	(0.044)	(0.044)	(0.043)
Std. O*NET Business skills^2	0.025***	0.023***	0.023***	0.025***	0.026***	0.027***	0.028***	0.028***
	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)
Std. O*NET Specialized skills	-0.128***	-0.140***	-0.134***	-0.136***	-0.135***	-0.134***	-0.132***	-0.120***
	(0.022)	(0.023)	(0.023)	(0.023)	(0.023)	(0.023)	(0.023)	(0.023)
Std. O*NET Specialized skills^2	-0.019**	-0.009	-0.011	-0.011	-0.012	-0.012	-0.012	-0.013*
	(0.008)	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)
Std. O*NET Balanced skill count	0.053***	0.042***	0.024	0.020	0.021	0.020	0.018	0.009
	(0.015)	(0.014)	(0.015)	(0.015)	(0.015)	(0.015)	(0.015)	(0.015)
Std. O*NET Balanced skill count^2	-0.001	-0.002	-0.003	-0.004	-0.004	-0.004	-0.004	-0.005
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Advanced Business Ownership (Unconditional Prob. = 0.004)								
Std. O*NET Business skills	0.425***	0.427***	0.494***	0.403***	0.373***	0.351***	0.344***	0.234***
	(0.065)	(0.065)	(0.065)	(0.068)	(0.069)	(0.069)	(0.069)	(0.067)
Std. O*NET Business skills^2	-0.014	-0.016	-0.024**	-0.016	-0.017	-0.015	-0.013	-0.008
	(0.010)	(0.011)	(0.011)	(0.011)	(0.011)	(0.011)	(0.011)	(0.011)
Std. O*NET Specialized skills	-0.074**	-0.087***	-0.044	-0.047	-0.045	-0.050	-0.048	-0.028
	(0.032)	(0.033)	(0.033)	(0.033)	(0.033)	(0.033)	(0.033)	(0.035)
Std. O*NET Specialized skills^2	-0.045***	-0.036***	-0.045***	-0.046***	-0.048***	-0.046***	-0.047***	-0.046***
	(0.011)	(0.011)	(0.012)	(0.011)	(0.011)	(0.011)	(0.011)	(0.011)
Std. O*NET Balanced skill count	0.124***	0.109***	0.015	0.013	0.013	0.015	0.010	0.003
	(0.026)	(0.026)	(0.027)	(0.026)	(0.026)	(0.026)	(0.026)	(0.026)
Std. O*NET Balanced skill count^2	0.009	0.008	-0.002	-0.002	-0.002	-0.001	-0.002	-0.001
	(0.005)	(0.005)	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)
Potential and actual experience	✓	✓	✓	✓	✓	✓	✓	✓
Past business ownership spells		✓	✓	✓	✓	✓	✓	✓
Gender, race, and household composition			✓	✓	✓	✓	✓	✓
Education and AFQT				✓	✓	✓	✓	✓
Wealth, income, and father's background					✓	✓	✓	✓
Risk tolerance and personality						✓	✓	✓
Family member owned a business							✓	✓
Considers oneself an entrepreneur								✓
Number of individuals	7,116	7,116	7,116	7,116	7,116	7,116	7,116	7,116
Number of person-year observations	251,214	251,214	251,214	251,214	251,214	251,214	251,214	251,214

Note: The models are those in Table 8, with the same controls. All reported measures are standardized so that a one-unit change equals half a standard deviation. Standard errors clustered at the individual level are in parentheses. *p < .10; **p < .05; ***p < .01.

Table A4: Are Business Skills Reflecting General Skill Requirements?

	Dependent Variable: Entry into Business Ownership		
	(1)	(2)	(3)
Multinomial Logit			
Basic Business Ownership (Unconditional Prob. = 0.014)			
Std. O*NET Business skills	-0.144***		-0.128***
	(0.043)		(0.047)
Std. O*NET Specialized skills	-0.131***		-0.127***
	(0.022)		(0.022)
Std. O*NET Balanced skill count	0.018		0.019
	(0.015)		(0.015)
Std. Actual experience in Job Zone 4+ occupations		-0.071***	-0.030
		(0.025)	(0.029)
Advanced Business Ownership (Unconditional Prob. = 0.004)			
Std. O*NET Business skills	0.344***		0.353***
	(0.068)		(0.073)
Std. O*NET Specialized skills	-0.048		-0.047
	(0.032)		(0.033)
Std. O*NET Balanced skill count	0.010		0.012
	(0.026)		(0.026)
Std. Actual experience in Job Zone 4+ occupations		0.090***	-0.019
		(0.035)	(0.040)
Number of individuals	7,116	7,116	7,116
Number of person-year observations	251,214	251,214	251,214

Note: Column 1 repeats column 5 of Table 7. Job Zone 4+ occupations are those in O*NET Job Zones 4 and 5, which require the most preparation. The reference category is staying out of business ownership. Estimated marginal effects are the average percentage change in the predicted probability of entry from a half-standard-deviation increase in each measure. See Table 1 for the full list of covariates included. Standard errors clustered at the individual level are in parentheses. *p < .10; **p < .05; ***p < .01.

Table A5: Tobit and OLS Coefficient Estimates for Business Revenue and Weekly Earnings in Business Ownership

	Revenue Sample		Earnings Sample
	Log Annual Revenue (Tobit)	Log Weekly Earnings (OLS)	Log Weekly Earnings (OLS)
	(1)	(2)	(3)
Panel A: Basic Business Ownership			
Std. O*NET Business skills	0.240**	0.125	0.027
	(0.103)	(0.102)	(0.063)
Std. O*NET Business skills^2	-0.030**	-0.031**	-0.005
	(0.015)	(0.014)	(0.011)
Std. O*NET Specialized skills	0.160***	0.140***	0.174***
	(0.046)	(0.052)	(0.035)
Std. O*NET Specialized skills^2	-0.007	0.003	-0.004
	(0.011)	(0.011)	(0.011)
Std. O*NET Balanced skill count	0.076*	0.045	0.036**
	(0.044)	(0.045)	(0.018)
Std. O*NET Balanced skill count^2	0.013**	0.006	0.002
	(0.006)	(0.006)	(0.002)
Tenure in the business	0.194***	0.167***	0.113***
	(0.020)	(0.021)	(0.010)
Tenure in the business^2	-0.004***	-0.003***	-0.003***
	(0.001)	(0.001)	(0.000)
Number of person-year observations	2,430	2,430	10,383
Number of businesses	1,221	1,221	4,118
Adjusted R^2 (pseudo R^2 for Tobit)	0.061	0.179	0.187
Mean business revenue or earnings	56,859	412	514
Panel B: Advanced Business Ownership			
Std. O*NET Business skills	0.439***	0.107	0.104
	(0.120)	(0.097)	(0.072)
Std. O*NET Business skills^2	-0.050***	-0.032**	-0.031***
	(0.017)	(0.015)	(0.012)
Std. O*NET Specialized skills	0.065	0.052	0.078*
	(0.071)	(0.063)	(0.044)
Std. O*NET Specialized skills^2	0.007	0.031**	0.022**
	(0.014)	(0.013)	(0.010)
Std. O*NET Balanced skill count	0.007	0.039	0.035
	(0.058)	(0.042)	(0.029)
Std. O*NET Balanced skill count^2	0.009	0.008	0.004
	(0.009)	(0.007)	(0.003)
Tenure in the business	0.175***	0.188***	0.134***
	(0.022)	(0.019)	(0.011)
Tenure in the business^2	-0.003***	-0.004***	-0.003***
	(0.001)	(0.001)	(0.000)
Number of person-year observations	2,465	2,465	5,938
Number of businesses	1,027	1,027	1,554
Adjusted R^2 (pseudo R^2 for Tobit)	0.065	0.177	0.167
Mean business revenue or earnings	215,578	1,088	1,088

Note: The models are those in Table 9, with the same controls. Skill measures are standardized with the mean and standard deviation of the revenue sample so that a one-unit increase equals half a standard deviation. Mean revenue in column 1 excludes top-coded reports. Standard errors clustered at the individual level are in parentheses. *p < .10; **p < .05; ***p < .01.

Appendix B: Other Approaches to the Endogeneity Problem

My estimates rely on selection on observables. Two other approaches could address the endogeneity problem discussed in Section 5.3 without requiring me to observe the tastes that create it: individual fixed effects and an instrument for skills. In this appendix, I describe what each approach would address and why I rely on selection on observables in the main analysis instead.

The error term ϵ_{ijt} in equation (11) captures factors that shift the utility of option j but are not included in the observed regressors. Because BX_{it} and SX_{it} are built from workers' occupational histories, they depend not only on the skills each occupation uses but also on workers' choices of which occupations to enter and how long to stay in them. The main specification controls for experience, demographic characteristics, wealth, risk tolerance, cognitive ability, personality traits, and family background, but tastes I do not observe may still shape how workers allocate their years across occupations.

To show how unobserved tastes may affect the estimates, decompose ϵ_{ijt} as $\epsilon_{ijt} = u_{ij} + v_{ijt}$, where u_{ij} is time-invariant and v_{ijt} varies within a worker across years. The time-invariant component u_{ij} captures tastes that do not change over a worker's career, such as a desire for autonomy or an aspiration to own a business. A worker with such tastes may sort into occupations that use more business skills and stay in them longer in anticipation of entry. This creates a positive correlation between BX_{it} and u_{ij} for the business ownership options, so the estimated association between business skills and entry may reflect both the skills themselves and these tastes. The same issue applies to SX_{it}.

The time-varying component v_{ijt} captures changes in a worker's plans or circumstances over the career. For example, a worker who becomes interested in business ownership in mid-career may move into an occupation that uses more business skills before entering. Within-person changes in skills are then correlated with unobserved changes in the attractiveness of business ownership. Studies of returns to occupational tenure (Kambourov and Manovskii, 2009) and task-specific human capital (Gathmann and Schönberg, 2010) face the same problem. In these studies, workers stay longer in occupations that match them well, and because match quality changes when workers switch occupations, individual fixed effects cannot remove it.

B.1 Individual Fixed Effects for Entry into Business Ownership

Individual fixed effects would address the time-invariant component u_{ij}. The relationship between skills and entry into business ownership would be identified from within-person changes in BX_{it} and SX_{it} as workers move across occupations, so time-invariant tastes would no longer conflate the estimates. Because the entry model is a MNL, this requires

a conditional logit model following Chamberlain (1980). This approach avoids imposing a distributional assumption on the individual effects and allows u_{ij} to be arbitrarily correlated with the observed covariates. A conditional rather than unconditional approach is necessary because estimating individual-specific intercepts in a nonlinear model introduces the incidental parameters problem.

However, individual fixed effects do not address the time-varying component v_{ijt}. A worker who becomes interested in business ownership in mid-career and then moves into occupations that use more business skills still generate within-person variation in BX_{it} that is correlated with her changing plans.

Moreover, individual fixed effects are less informative in my entry model. Because BX_{it} and SX_{it} accumulate steadily as a worker gains experience, little within-person variation in skills remains once experience is controlled for. Fixed effects are more useful for earnings, which are observed in both wage employment and business ownership. For earnings, the within-person variation comes mainly from moves between wage employment and business ownership rather than from changes in skills. The fixed effect absorbs a worker's overall earnings level, and the change in her earnings when she moves, compared across workers with different skills, shows how much more or less each skill pays in business ownership than in wage employment. I use this approach in Appendix B.2.

B.2 Individual Fixed Effects for Earnings

Because weekly earnings are observed in both wage employment and business ownership, unlike business revenue, I can compare the returns to business skills and specialized skills in business ownership with their returns in wage employment for the same workers. My argument about the returns to different skills in business ownership and wage employment (Section 3.2) also rests on this comparison. I argue that business skills are rewarded more in advanced business ownership than in wage employment and less in basic business ownership, and that wage employment is likely to reward specialized skills more than either type of business ownership (Predictions 1 and 2). I use these estimated returns to check whether the entry choices reported in Section 6.1 are consistent with how much each type of skill pays.

To make this comparison, I combine each worker's jobs in wage employment and in business ownership in one sample. I start with the earnings sample described in Section 4.2, which contains one observation for each business and year with reported weekly earnings. I then add one observation for each wage job and year with reported weekly earnings for the 7,116 individuals in the entry sample. The resulting sample contains 191,590 person-year observations, of which 87,251 come from the 2,976 individuals who ever own a business.

I estimate the returns to skills in wage employment and in each type of business ownership with the following model

$$\ln w_{it} = \alpha_i + H'_{it}\pi_0 + \sum_{j\in\{B,A\}} D^j_{it}\left(\delta_j + H'_{it}\pi_j\right) + W'_{it}\beta + \epsilon_{it}, \tag{B1}$$

where $\ln w_{it}$ is log weekly earnings and α_i is the individual fixed effect. D^B_{it} and D^A_{it} equal one when the job is a basic or an advanced business, so wage employment is the omitted category. H_{it} contains BX_{it} and SX_{it}, their squares, their interactions with $Tenure_{it}$, $Tenure_{it}$ and its square, and actual and potential experience and their squares. $Tenure_{it}$ is tenure in the current job, whether a wage job or a business. All of these variables are measured at the interview date. π_0 captures how the skill and experience terms are related to earnings in wage employment, and π_B and π_A capture how these relationships differ in basic and advanced business ownership. Because the skill measures are interacted with tenure, the return to each skill can change as a worker stays longer in her job. W_{it} contains the time-varying controls of the revenue and earnings models: marital and cohabitation status, the young-child indicator, the number of past business ownership spells, and region and industry fixed effects. The individual fixed effect absorbs the remaining controls, which change little or not at all over a career, such as education. See Table 1 for the full list of controls. Standard errors are clustered at the individual level.

Table B1 reports four versions of equation (B1), which move step by step from a comparison across all workers to a comparison within each worker. Column 1 uses all workers and, instead of the individual fixed effect, includes the time-invariant controls. Column 2 keeps only workers who ever own a business, so that earnings in wage employment come from the same workers whose earnings in business ownership are observed. Column 3 adds the individual fixed effect, which absorbs time-invariant individual characteristics. Because a worker reports her earnings at every interview while she holds a job, businesses that last longer contribute more observations. To keep these businesses from dominating the estimates, column 4 repeats column 3 but weights each observation by the inverse of the number of observations in its job spell, so that each spell counts equally.

To interpret the estimates in column 4 of Table B1, I calculate the average marginal effect of a half-standard-deviation increase in each skill measure, reported in Table B2. I do this separately for the two types of business ownership, using the observations of basic business owners in columns 1 to 3 and those of advanced business owners in columns 4 to 6. For each of these observations, I compute the return to a skill twice, once in business ownership and once as if the owner were a wage worker with the same skills and tenure, and then average each over the observations. The difference between the two averages shows how much more

or less a skill pays the same workers in business ownership than in wage employment. The table also reports these returns at zero, five, and ten years of tenure in the current job. I convert each estimate b to a percent change, $100(e^b - 1)$.

Business skills pay much less in basic business ownership than in wage employment (Prediction 1), whereas in advanced business ownership the difference is small. In columns 1 and 2 of Table B2, a half-standard-deviation increase in business skills predicts 7.0 percent higher earnings for basic business owners in wage employment but 16.2 percent lower earnings in basic business ownership. The difference in column 3 is statistically significant at every level of tenure, although it narrows as owners stay in their businesses, and the return in business ownership rises from -17.3 percent at the start of a business to -9.0 percent after ten years. In columns 4 and 5, the same increase is associated with 5.9 percent higher earnings in wage employment and 0.3 percent higher earnings in advanced business ownership. The difference in column 6 is not statistically significant at any level of tenure. The return in advanced business ownership also grows with tenure, from -2.7 percent at the start of a business to 5.0 percent after ten years, close to the 6.4 percent return in wage employment. These estimates do not show that business skills pay more in advanced business ownership than in wage employment. Still, compared with wage employment, they pay much more in advanced than in basic business ownership. As discussed in Section 6.2, a possible reason is that owners with more business skills reinvest profits in their businesses rather than taking them home as earnings, which would lead earnings to understate the return to business skills in advanced business ownership.

The estimates do not support the prediction that specialized skills pay more in wage employment than in business ownership (Prediction 2), because the differences in columns 3 and 6 have the opposite sign, although none of them is statistically significant. In columns 1 and 2 of Table B2, a half-standard-deviation increase in specialized skills predicts 6.0 percent higher earnings for basic business owners in business ownership and 3.8 percent higher earnings for the same owners in wage employment. For advanced business owners, the same increase is associated with 7.0 percent higher earnings in business ownership and 2.8 percent higher earnings in wage employment.

Comparing columns 1 to 3 of Table B1 also shows whether these differences in returns are driven by selection on time-invariant individual characteristics. This selection is part of the endogeneity problem discussed in Section 5.3. The estimated coefficient on the interaction between business skills and basic business ownership changes little, from -0.298 in column 1 to -0.238 in column 3. Restricting the sample to workers who ever own a business and adding the individual fixed effect change it only slightly, so selection on time-invariant individual characteristics does not explain the low return to business skills in basic business ownership.

In contrast, the estimated coefficient for advanced business ownership shrinks by more than a quarter, from -0.135 in column 1 to -0.097 in column 3, which might reflect some selection on these characteristics in columns 1 and 2. The estimated coefficients on the interactions of specialized skills with both types of business ownership are positive in all three columns and become slightly smaller as the sample is restricted and the individual fixed effect is added.

Table B1: Coefficient Estimates for Log Weekly Earnings using Individual Fixed Effects

	Dependent Variable: Log Weekly Earnings			
	(1)	(2)	(3)	(4)
Std. O*NET Business skills	0.153***	0.152***	0.058***	0.066***
	(0.011)	(0.018)	(0.017)	(0.021)
Std. O*NET Business skills^2	-0.010***	-0.010***	-0.003**	-0.004
	(0.001)	(0.002)	(0.002)	(0.003)
Std. O*NET Business skills × tenure in current job	0.002***	0.002***	0.003***	0.001
	(0.000)	(0.001)	(0.001)	(0.001)
Std. O*NET Business skills × 1 if basic business owner	-0.298***	-0.273***	-0.238***	-0.257***
	(0.055)	(0.055)	(0.053)	(0.059)
Std. O*NET Business skills^2 × 1 if basic business owner	0.019***	0.016**	0.011*	0.012*
	(0.007)	(0.007)	(0.006)	(0.007)
Std. O*NET Business skills × tenure in current job × 1 if basic business owner	0.005	0.004	0.005*	0.009***
	(0.003)	(0.003)	(0.003)	(0.003)
Std. O*NET Business skills × 1 if advanced business owner	-0.135**	-0.119**	-0.097*	-0.045
	(0.057)	(0.058)	(0.059)	(0.080)
Std. O*NET Business skills^2 × 1 if advanced business owner	-0.004	-0.006	-0.006	-0.012
	(0.007)	(0.007)	(0.007)	(0.009)
Std. O*NET Business skills × tenure in current job × 1 if advanced business owner	0.007***	0.007***	0.006**	0.007**
	(0.003)	(0.003)	(0.003)	(0.003)
Std. O*NET Specialized skills	0.043***	0.051***	0.028**	0.040***
	(0.006)	(0.011)	(0.011)	(0.013)
Std. O*NET Specialized skills^2	0.001	-0.001	-0.001	-0.002
	(0.001)	(0.002)	(0.001)	(0.002)
Std. O*NET Specialized skills × tenure in current job	-0.002***	-0.001**	-0.001**	-0.002**
	(0.000)	(0.001)	(0.001)	(0.001)
Std. O*NET Specialized skills × 1 if basic business owner	0.057*	0.050	0.039	0.017
	(0.033)	(0.033)	(0.032)	(0.037)
Std. O*NET Specialized skills^2 × 1 if basic business owner	0.002	0.004	0.005	0.005
	(0.007)	(0.007)	(0.006)	(0.006)
Std. O*NET Specialized skills × tenure in current job × 1 if basic business owner	0.002	0.002	0.001	0.002
	(0.002)	(0.003)	(0.002)	(0.003)
Std. O*NET Specialized skills × 1 if advanced business owner	0.046	0.036	0.034	0.015
	(0.043)	(0.043)	(0.042)	(0.053)
Std. O*NET Specialized skills^2 × 1 if advanced business owner	0.007	0.009*	0.007	0.013**
	(0.005)	(0.005)	(0.005)	(0.006)
Std. O*NET Specialized skills × tenure in current job × 1 if advanced business owner	-0.004*	-0.005*	-0.003	-0.003
	(0.002)	(0.002)	(0.002)	(0.003)
1 if basic business owner	-0.466***	-0.398***	-0.354***	-0.447***
	(0.133)	(0.132)	(0.129)	(0.149)
1 if advanced business owner	0.568***	0.609***	0.530***	0.675***
	(0.154)	(0.154)	(0.158)	(0.201)
Tenure in current job	0.065***	0.074***	0.075***	0.086***
	(0.001)	(0.002)	(0.002)	(0.003)
Tenure in current job^2	-0.002***	-0.002***	-0.002***	-0.002***
	(0.000)	(0.000)	(0.000)	(0.000)
Tenure in current job × 1 if basic business owner	0.096***	0.087***	0.072***	0.076***
	(0.011)	(0.011)	(0.011)	(0.013)
Tenure in current job^2 × 1 if basic business owner	-0.003***	-0.002***	-0.002***	-0.003***

Continued on next page

Table B1 (continued)

	Dependent Variable: Log Weekly Earnings			
	(1)	(2)	(3)	(4)
	(0.001)	(0.001)	(0.001)	(0.001)
Tenure in current job × 1 if advanced business owner	0.089***	0.080***	0.067***	0.078***
	(0.012)	(0.012)	(0.012)	(0.015)
Tenure in current job^2 × 1 if advanced business owner	-0.002***	-0.002***	-0.002***	-0.002***
	(0.000)	(0.000)	(0.000)	(0.001)
Actual experience	0.018***	0.024***	0.019***	0.017***
	(0.003)	(0.005)	(0.006)	(0.006)
Actual experience^2	-0.001***	-0.001***	-0.000***	-0.000***
	(0.000)	(0.000)	(0.000)	(0.000)
Actual experience × 1 if basic business owner	0.025	0.016	-0.002	0.001
	(0.016)	(0.017)	(0.015)	(0.018)
Actual experience^2 × 1 if basic business owner	-0.000	0.000	0.000	0.001
	(0.000)	(0.000)	(0.000)	(0.000)
Actual experience × 1 if advanced business owner	0.025	0.017	0.017	0.028
	(0.031)	(0.031)	(0.030)	(0.036)
Actual experience^2 × 1 if advanced business owner	-0.000	-0.000	-0.000	-0.001
	(0.001)	(0.001)	(0.001)	(0.001)
Potential experience	-0.026***	-0.031***	-0.008**	-0.006
	(0.002)	(0.003)	(0.004)	(0.004)
Potential experience^2	0.000***	0.000***	0.000	-0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Potential experience × 1 if basic business owner	-0.076***	-0.072***	-0.058***	-0.058***
	(0.012)	(0.012)	(0.011)	(0.012)
Potential experience^2 × 1 if basic business owner	0.000*	0.000*	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Potential experience × 1 if advanced business owner	-0.140***	-0.134***	-0.132***	-0.172***
	(0.028)	(0.028)	(0.027)	(0.030)
Potential experience^2 × 1 if advanced business owner	0.002***	0.002***	0.002***	0.003***
	(0.001)	(0.001)	(0.001)	(0.001)
Time-invariant baseline regressors	✓	✓		
Only workers who ever own a business		✓	✓	✓
Individual fixed effects			✓	✓
Weighted by inverse of spell-report frequency				✓
Mean dependent variable	6.157	6.055	6.055	6.055
Number of person-year observations	191,590	87,255	87,251	87,251
Adjusted R^2	0.322	0.292	0.382	0.364

Note: O*NET-based skill measures are standardized so that a one-unit increase equals half a standard deviation. See Table 1 for the full list of covariates included. Standard errors clustered at the individual level are in parentheses. *p < .10; **p < .05; ***p < .01.

Table B2: Estimated Returns to Business Skills and Specialized Skills for Business Owners, in Business Ownership and in Wage Employment (Based on Column 4 of Table B1)

	Basic Business Ownership			Advanced Business Ownership		
	Wage Employment (1)	Business Ownership (2)	Difference (2)−(1) (3)	Wage Employment (4)	Business Ownership (5)	Difference (5)−(4) (6)
Std. O*NET Business skills						
Average	0.067***	-0.177***	-0.245***	0.058***	0.003	-0.055
	(0.021)	(0.062)	(0.059)	(0.015)	(0.059)	(0.059)
Tenure 0	0.066***	-0.190***	-0.257***	0.054***	-0.028	-0.082
	(0.021)	(0.062)	(0.059)	(0.016)	(0.064)	(0.063)
Tenure 5	0.070***	-0.143**	-0.213***	0.058***	0.010	-0.048
	(0.020)	(0.063)	(0.060)	(0.015)	(0.059)	(0.058)
Tenure 10	0.074***	-0.095	-0.169***	0.062***	0.049	-0.014
	(0.020)	(0.067)	(0.065)	(0.015)	(0.058)	(0.058)
Std. O*NET Specialized skills						
Average	0.038***	0.059	0.021	0.027***	0.068*	0.040
	(0.013)	(0.037)	(0.036)	(0.009)	(0.039)	(0.039)
Tenure 0	0.040***	0.057	0.018	0.034***	0.085**	0.051
	(0.013)	(0.037)	(0.037)	(0.011)	(0.042)	(0.043)
Tenure 5	0.032***	0.061*	0.030	0.026***	0.063*	0.038
	(0.012)	(0.037)	(0.037)	(0.009)	(0.038)	(0.039)
Tenure 10	0.023**	0.065	0.042	0.018**	0.041	0.024
	(0.012)	(0.040)	(0.041)	(0.009)	(0.039)	(0.040)
Business owner-years		10,380			5,938	

Note: Estimated returns correspond to a half-standard-deviation increase in each skill and are in log points. In columns 1 and 4, the same business owners are treated as wage workers with the same skills and tenure. Average rows use each owner's observed tenure in the current job. Standard errors clustered at the individual level are in parentheses. *p < .10; **p < .05; ***p < .01.

B.3 Instrumental Variables

An instrument may address both components of the error term. An effective instrument is required to shift the skills a worker accumulates but affect business ownership only through those skills. One candidate is the industrial composition of the county where a worker starts her career, in the spirit of Jara-Figueroa et al. (2018). Different occupations offered because of the circumstances of the local labor markets may affect the skills that workers accumulate. For example, a county whose economy is concentrated in manufacturing exposes workers to a different set of occupations, and with them a different path of skill accumulation, than a county with a larger service sector or a concentration of professional industries. As a result, workers with similar initial characteristics may accumulate different skills because they begin their careers in different local labor markets.

I construct such an instrument for business skills, BX_{it}. For each county c, I use the 1975 County Business Patterns to calculate the share s_{kc} of employment in each industry k. I then measure how much each industry uses business skills, $\overline{BX}_k$, by assigning each occupation its business-skill score from Section 4.4 and averaging across workers employed in industry k in the 1980 Census. The instrument for worker i, whose county of residence at labor market entry is $c(i)$, is:

$$Z_{ict}^{BX} = \sum_k s_{kc(i),1975} \cdot g_{kt} \cdot \overline{BX}_k, \tag{B2}$$

where g_{kt} is national cumulative employment growth in industry k from 1975 to year t. The instrument is higher for workers who started their careers in counties where industries that use more business skills accounted for a large share of employment and later grew nationally.

I construct a similar instrument for specialized skills:

$$Z_{ict}^{SX} = \sum_k s_{kc(i),1975} \cdot \overline{SX}_{kt}. \tag{B3}$$

Because specialized skills depend on each worker's target occupation, the industry score $\overline{SX}_{kt}$ cannot simply average occupation-level scores. Instead, it measures how much the occupations within industry k share their most important skills. For instance, in the construction industry, plumbers, electricians, and carpenters rely on many of the same skills, so working in any of these occupations builds skills that are useful in the others. In hospitals, nurses, accountants, and janitors share few of their most important skills. Workers who start their careers in counties dominated by industries of the first kind accumulate more transferable specialized skills. The score changes over time as the occupational mix within each industry changes, which I measure at each Census and American Community Survey benchmark from 1980 to 2015.

For either instrument to be valid, the industrial composition of a worker's starting county must affect business ownership only through the skills she accumulates. Two concerns remain. First, local economic conditions may directly affect business ownership decisions. For example, limited wage opportunities may push workers into business ownership regardless of their skills. Controlling for the county unemployment rate at labor market entry would absorb part of this channel. Second, workers may sort across counties based on their tastes for business ownership. Using the county of residence at labor market entry, rather than at the time of the business ownership decision, reduces this concern, as does restricting the sample to workers who stay in the same county during the period used to construct the instrument.

However, I face two problems in applying these two instruments. First, they are highly correlated, with a correlation of about 0.9 across counties. Industries whose occupations share business skills, such as finance and professional services, also rank high on how much their occupations share specialized skills. As a result, the instruments cannot separate the two types of skills, which the entry model requires. Second, the business-skill instrument is weak in a first stage with individual fixed effects, the setting of Appendix B.2, where its only source of variation within a worker is national industry growth over time. Separating business from specialized skills would require a source of variation that shifts one type of skill without shifting the other, which I leave for future work.